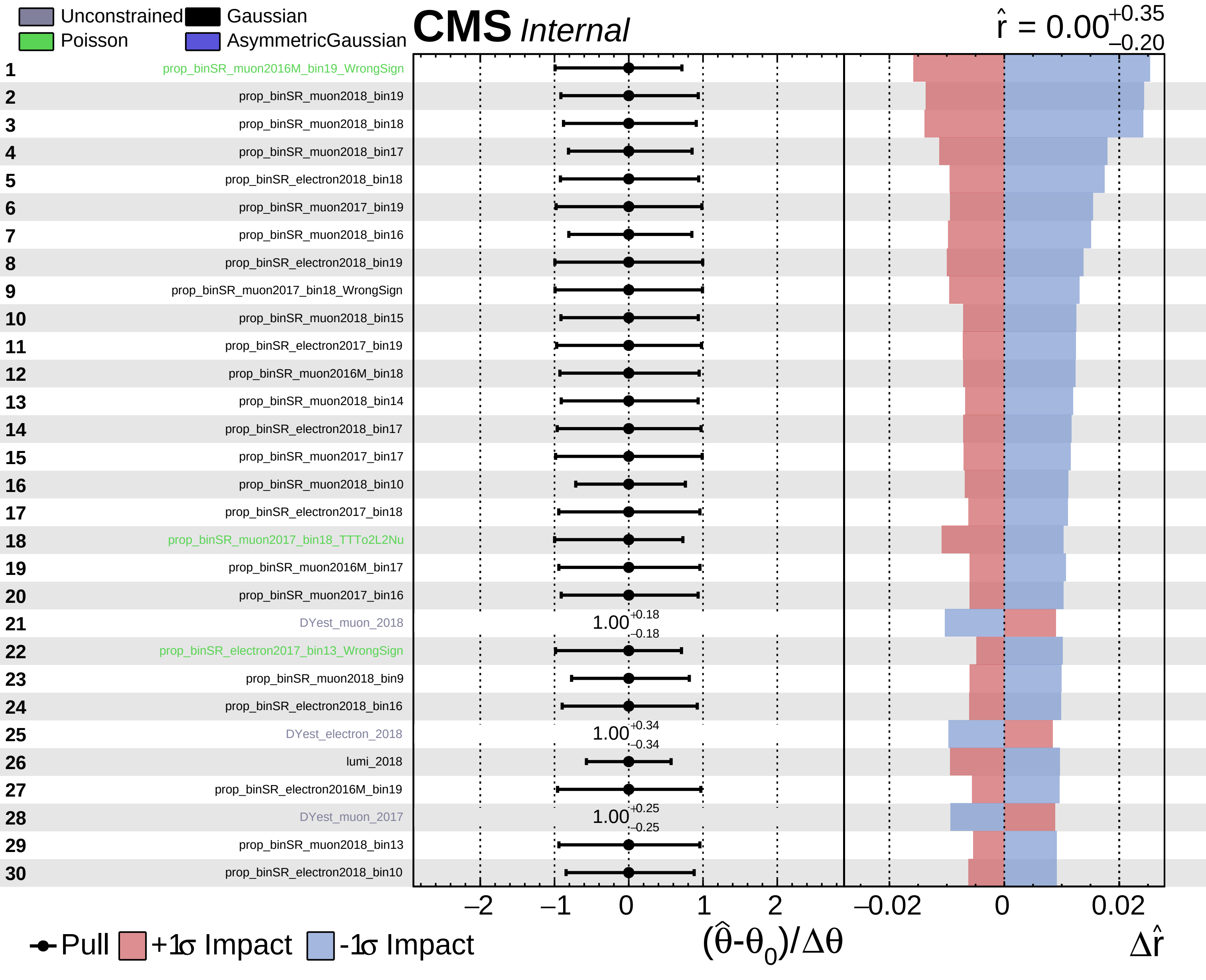
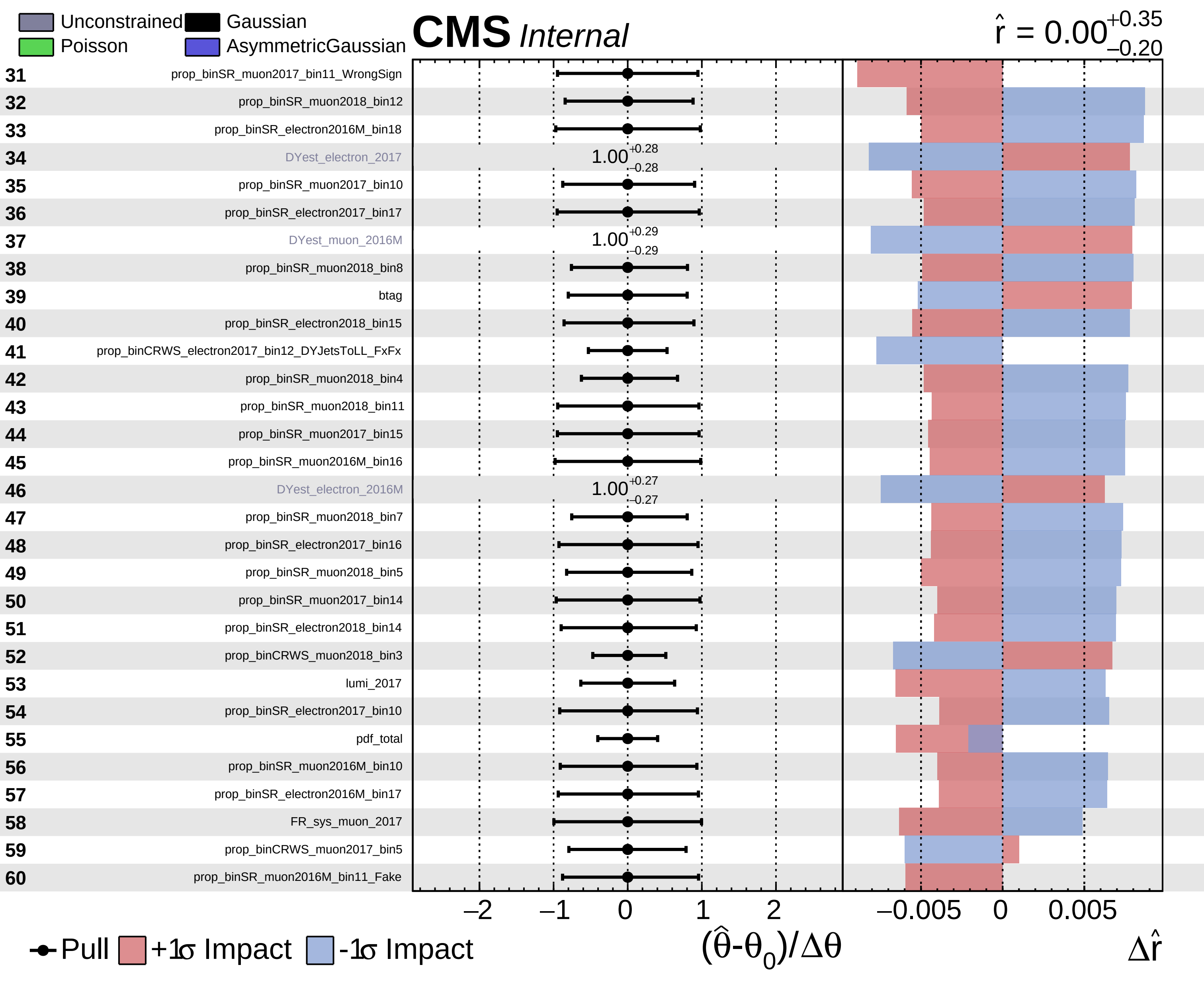
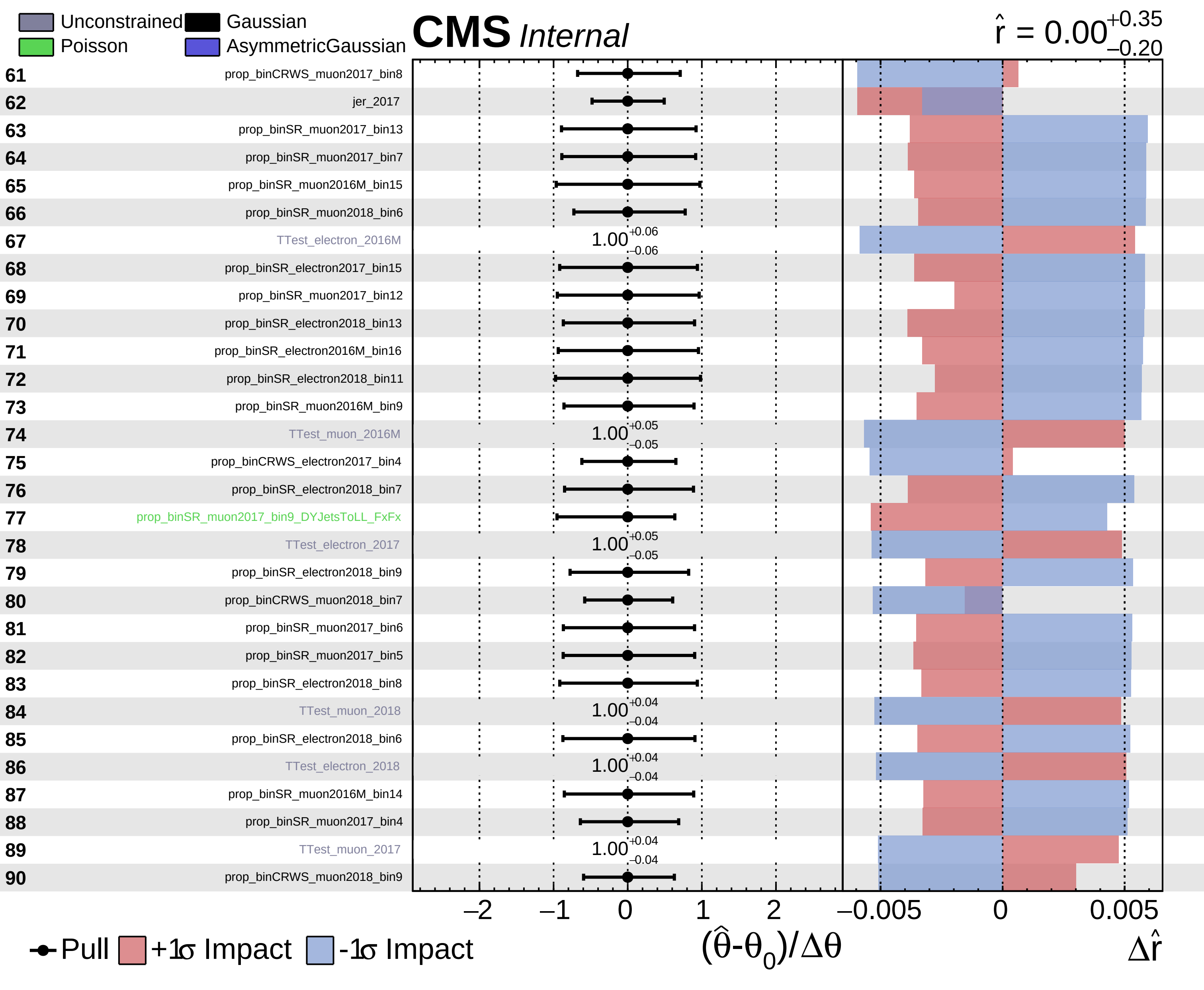
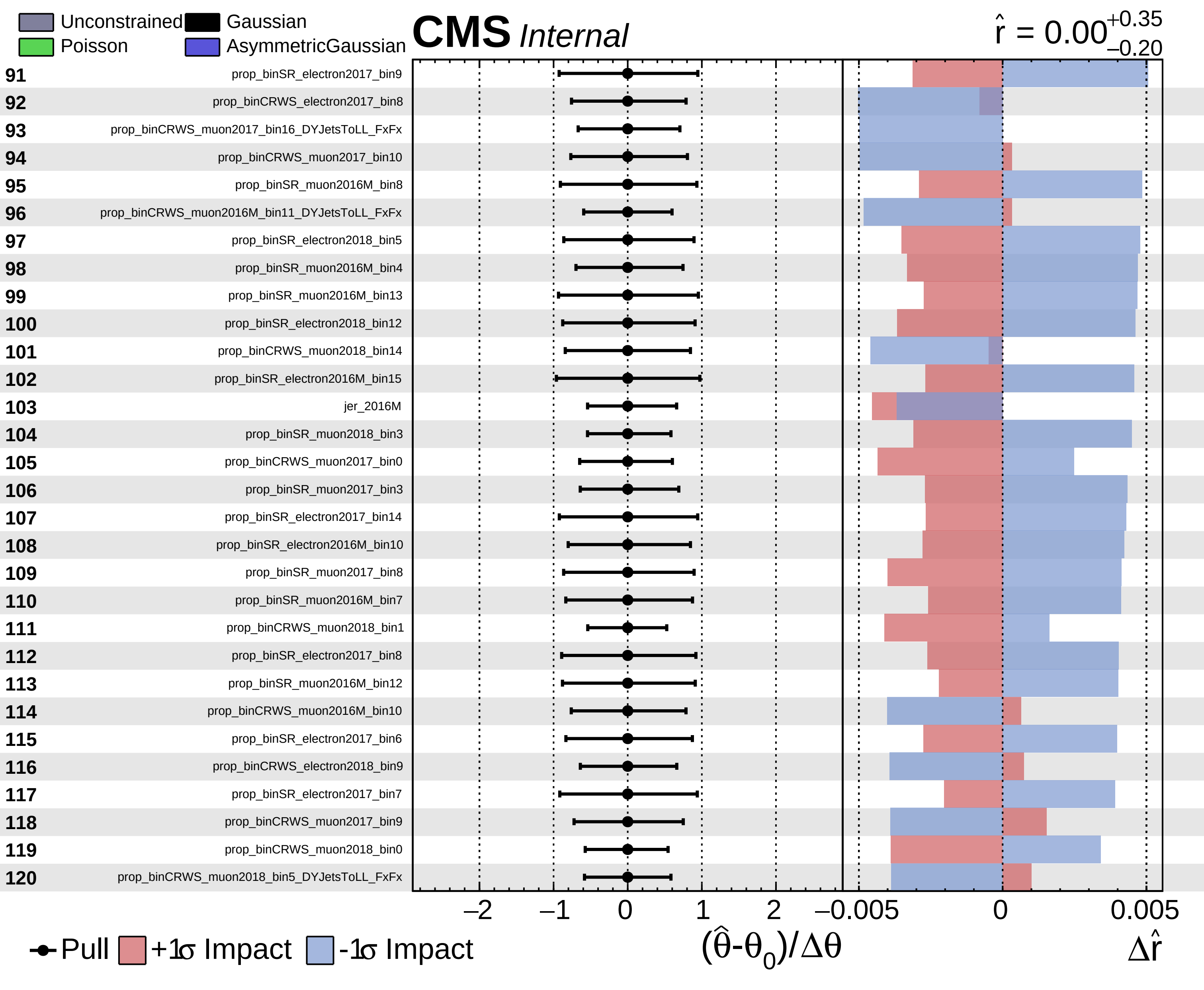


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
1 prop_binSR_muon2016M_bin19_WrongSign
2 prop_binSR_muon2018_bin19
3 prop_binSR_muon2018_bin18
4 prop_binSR_muon2018_bin17
5 prop_binSR_electron2018_bin18
6 prop_binSR_muon2017_bin19
7 prop_binSR_muon2018_bin16
8 prop_binSR_electron2018_bin19
9 prop_binSR_muon2017_bin18_WrongSign
10 prop_binSR_muon2018_bin15
11 prop_binSR_electron2017_bin19
12 prop_binSR_muon2016M_bin18
13 prop_binSR_muon2018_bin14
14 prop_binSR_electron2018_bin17
15 prop_binSR_muon2017_bin17
16 prop_binSR_muon2018_bin10
17 prop_binSR_electron2017_bin18
18 prop_binSR_muon2017_bin18_TTTo2L2Nu
19 prop_binSR_muon2016M_bin17
20 prop_binSR_muon2017_bin16
21 DYest_muon_2018
$1.00^{+0.18}_{-0.18}$
22 prop_binSR_electron2017_bin13_WrongSign
23 prop_binSR_muon2018_bin9
24 prop_binSR_electron2018_bin16
25 DYest_electron_2018
$1.00^{+0.34}_{-0.34}$
26 lumi_2018
27 prop_binSR_electron2016M_bin19
28 DYest_muon_2017
$1.00^{+0.25}_{-0.25}$
29 prop_binSR_muon2018_bin13
30 prop_binSR_electron2018_bin10
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

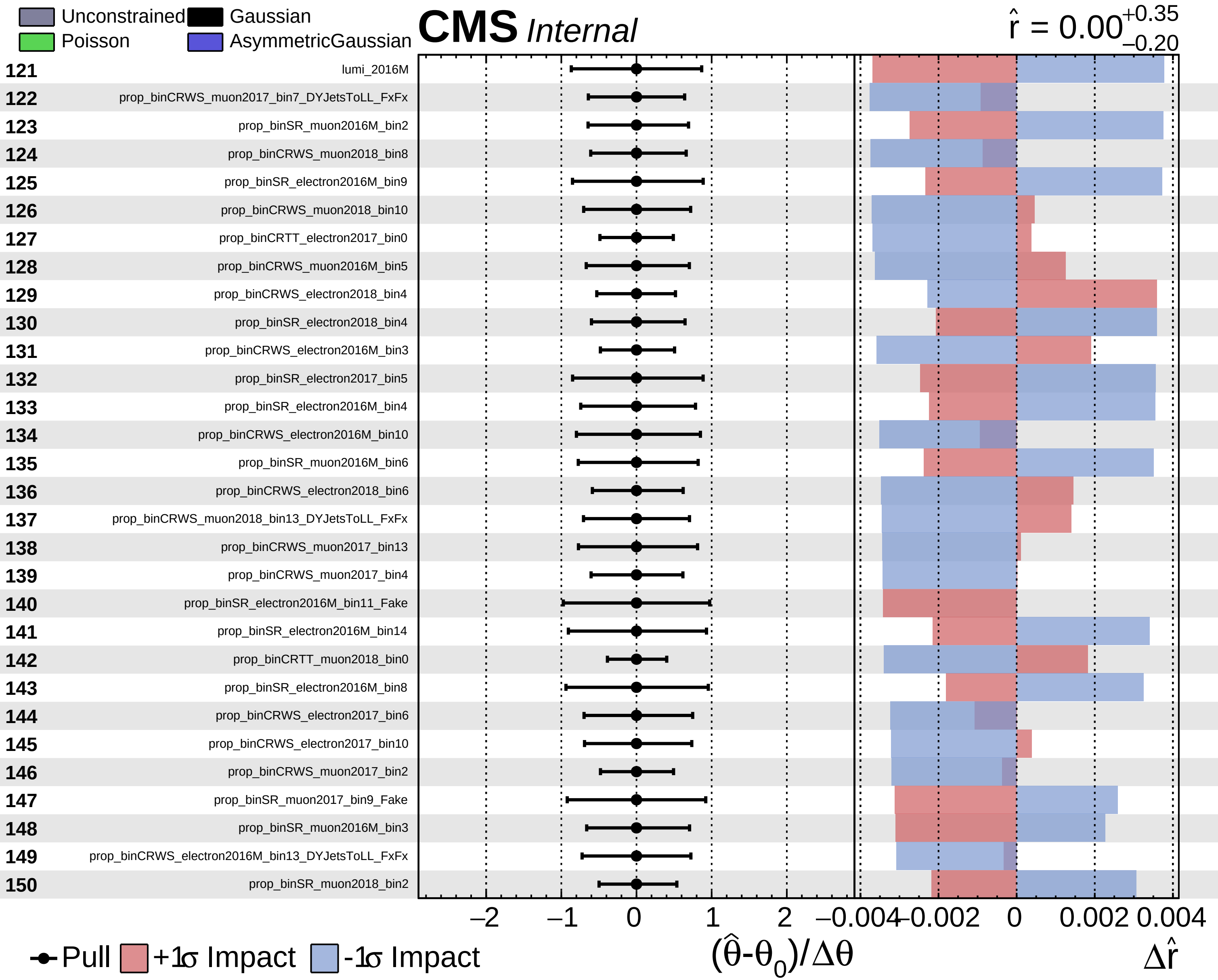

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.35 −0.20
31 prop_binSR_muon2017_bin11_WrongSign
32 prop_binSR_muon2018_bin12
33 prop_binSR_electron2016M_bin18
34 DYest_electron_2017
1.00 +0.28 −0.28
35 prop_binSR_muon2017_bin10
36 prop_binSR_electron2017_bin17
37 DYest_muon_2016M
1.00 +0.29 −0.29
38 prop_binSR_muon2018_bin8
39 btag
40 prop_binSR_electron2018_bin15
41 prop_binCRWS_electron2017_bin12_DYJetsToLL_FxFx
42 prop_binSR_muon2018_bin4
43 prop_binSR_muon2018_bin11
44 prop_binSR_muon2017_bin15
45 prop_binSR_muon2016M_bin16
46 DYest_electron_2016M
1.00 +0.27 −0.27
47 prop_binSR_muon2018_bin7
48 prop_binSR_electron2017_bin16
49 prop_binSR_muon2018_bin5
50 prop_binSR_muon2017_bin14
51 prop_binSR_electron2018_bin14
52 prop_binCRWS_muon2018_bin3
53 lumi_2017
54 prop_binSR_electron2017_bin10
55 pdf_total
56 prop_binSR_muon2016M_bin10
57 prop_binSR_electron2016M_bin17
58 FR_sys_muon_2017
59 prop_binCRWS_muon2017_bin5
60 prop_binSR_muon2016M_bin11_Fake
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

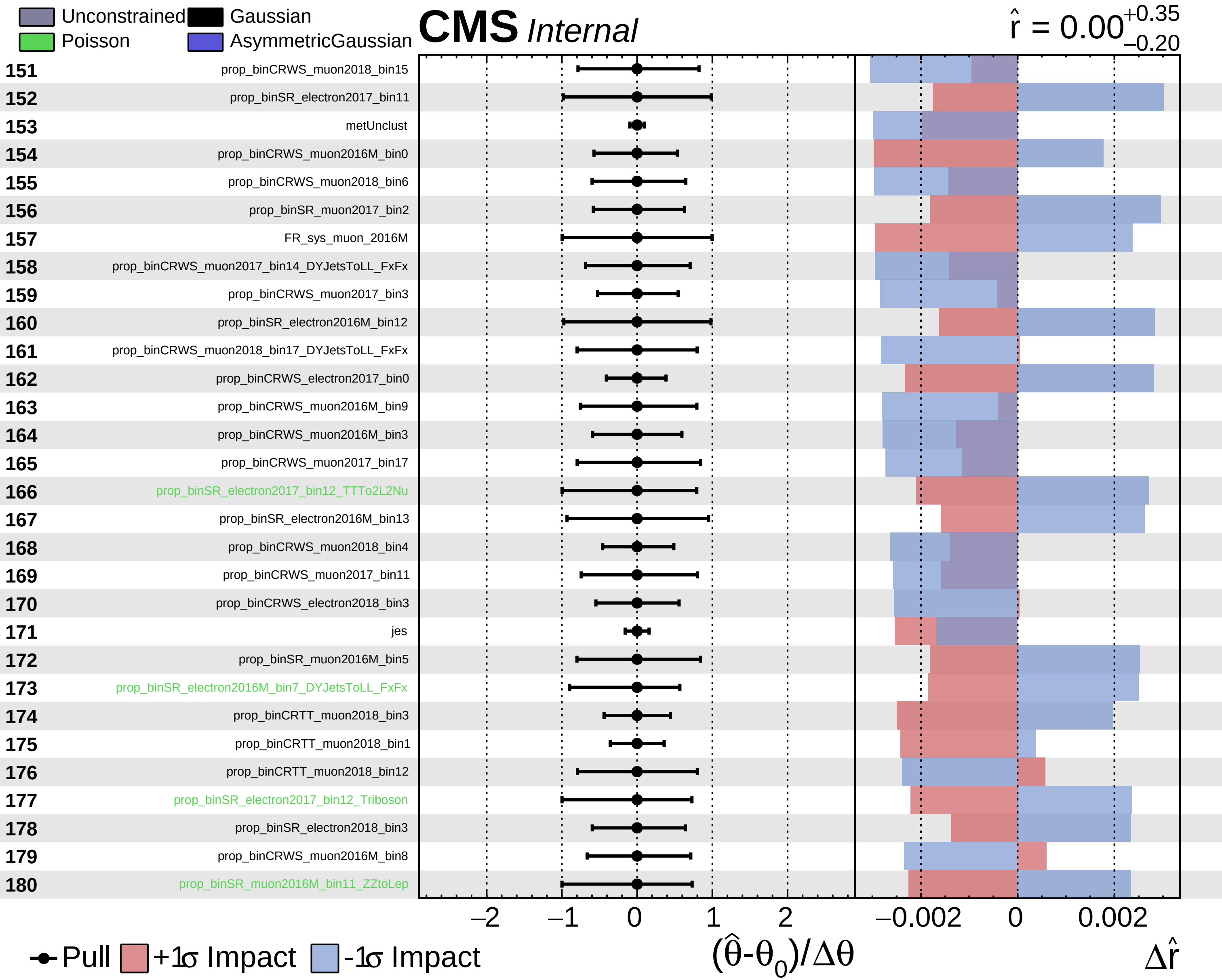


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
61 prop_binCRWS_muon2017_bin8
62 jer_2017
63 prop_binSR_muon2017_bin13
64 prop_binSR_muon2017_bin7
65 prop_binSR_muon2016M_bin15
66 prop_binSR_muon2018_bin6
67 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
68 prop_binSR_electron2017_bin15
69 prop_binSR_muon2017_bin12
70 prop_binSR_electron2018_bin13
71 prop_binSR_electron2016M_bin16
72 prop_binSR_electron2018_bin11
73 prop_binSR_muon2016M_bin9
74 TTest_muon_2016M
$1.00^{+0.05}_{-0.05}$
75 prop_binCRWS_electron2017_bin4
76 prop_binSR_electron2018_bin7
77 prop_binSR_muon2017_bin9_DYJetsToLL_FxFx
78 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
79 prop_binSR_electron2018_bin9
80 prop_binCRWS_muon2018_bin7
81 prop_binSR_muon2017_bin6
82 prop_binSR_muon2017_bin5
83 prop_binSR_electron2018_bin8
84 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
85 prop_binSR_electron2018_bin6
86 TTest_electron_2018
$1.00^{+0.04}_{-0.04}$
87 prop_binSR_muon2016M_bin14
88 prop_binSR_muon2017_bin4
89 TTest_muon_2017
$1.00^{+0.04}_{-0.04}$
90 prop_binCRWS_muon2018_bin9
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

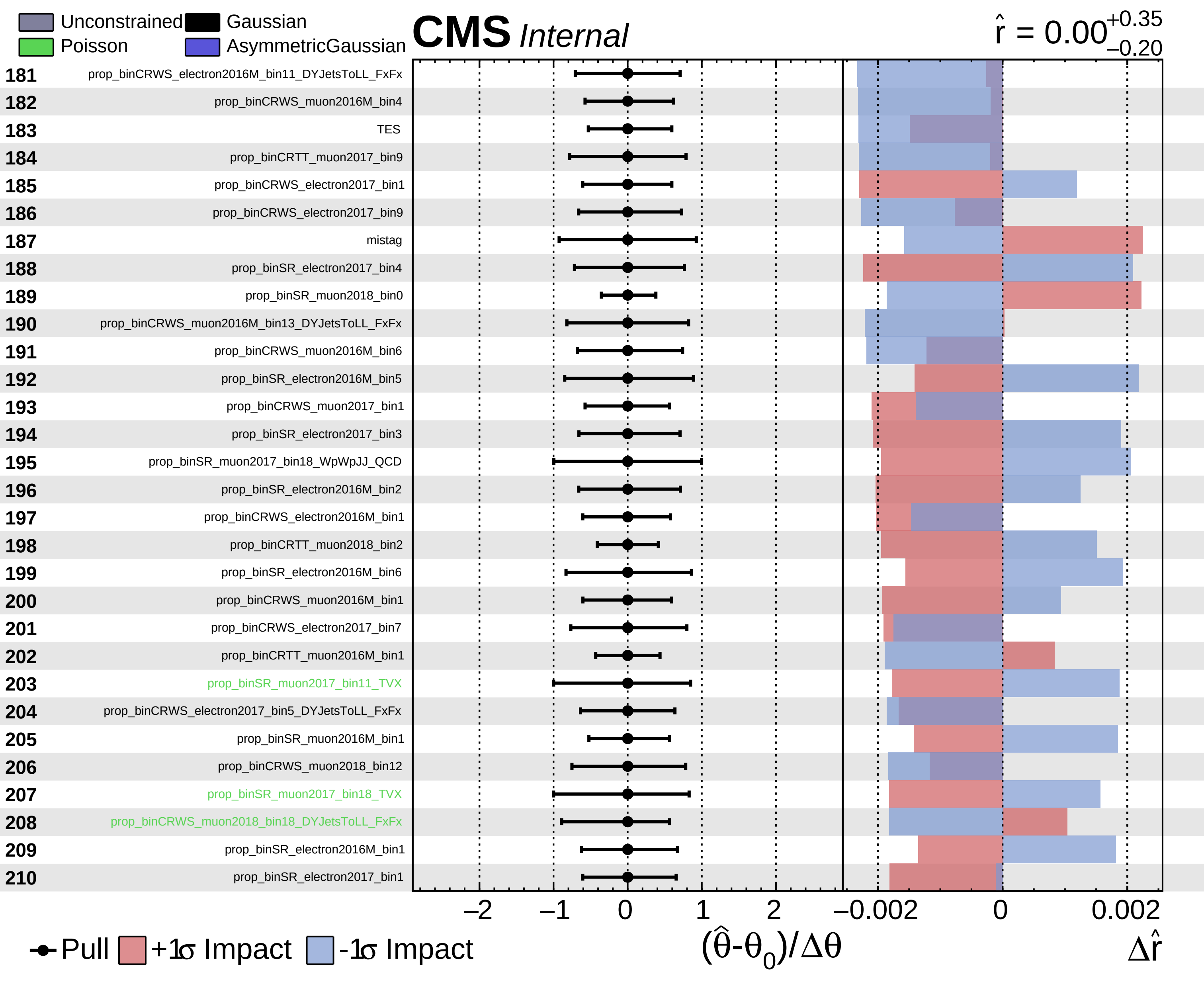


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
91 prop_binSR_electron2017_bin9
92 prop_binCRWS_electron2017_bin8
93 prop_binCRWS_muon2017_bin16_DYJetsToLL_FxFx
94 prop_binCRWS_muon2017_bin10
95 prop_binSR_muon2016M_bin8
96 prop_binCRWS_muon2016M_bin11_DYJetsToLL_FxFx
97 prop_binSR_electron2018_bin5
98 prop_binSR_muon2016M_bin4
99 prop_binSR_muon2016M_bin13
100 prop_binSR_electron2018_bin12
101 prop_binCRWS_muon2018_bin14
102 prop_binSR_electron2016M_bin15
103 jer_2016M
104 prop_binSR_muon2018_bin3
105 prop_binCRWS_muon2017_bin0
106 prop_binSR_muon2017_bin3
107 prop_binSR_electron2017_bin14
108 prop_binSR_electron2016M_bin10
109 prop_binSR_muon2017_bin8
110 prop_binSR_muon2016M_bin7
111 prop_binCRWS_muon2018_bin1
112 prop_binSR_electron2017_bin8
113 prop_binSR_muon2016M_bin12
114 prop_binCRWS_muon2016M_bin10
115 prop_binSR_electron2017_bin6
116 prop_binCRWS_electron2018_bin9
117 prop_binSR_electron2017_bin7
118 prop_binCRWS_muon2017_bin9
119 prop_binCRWS_muon2018_bin0
120 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

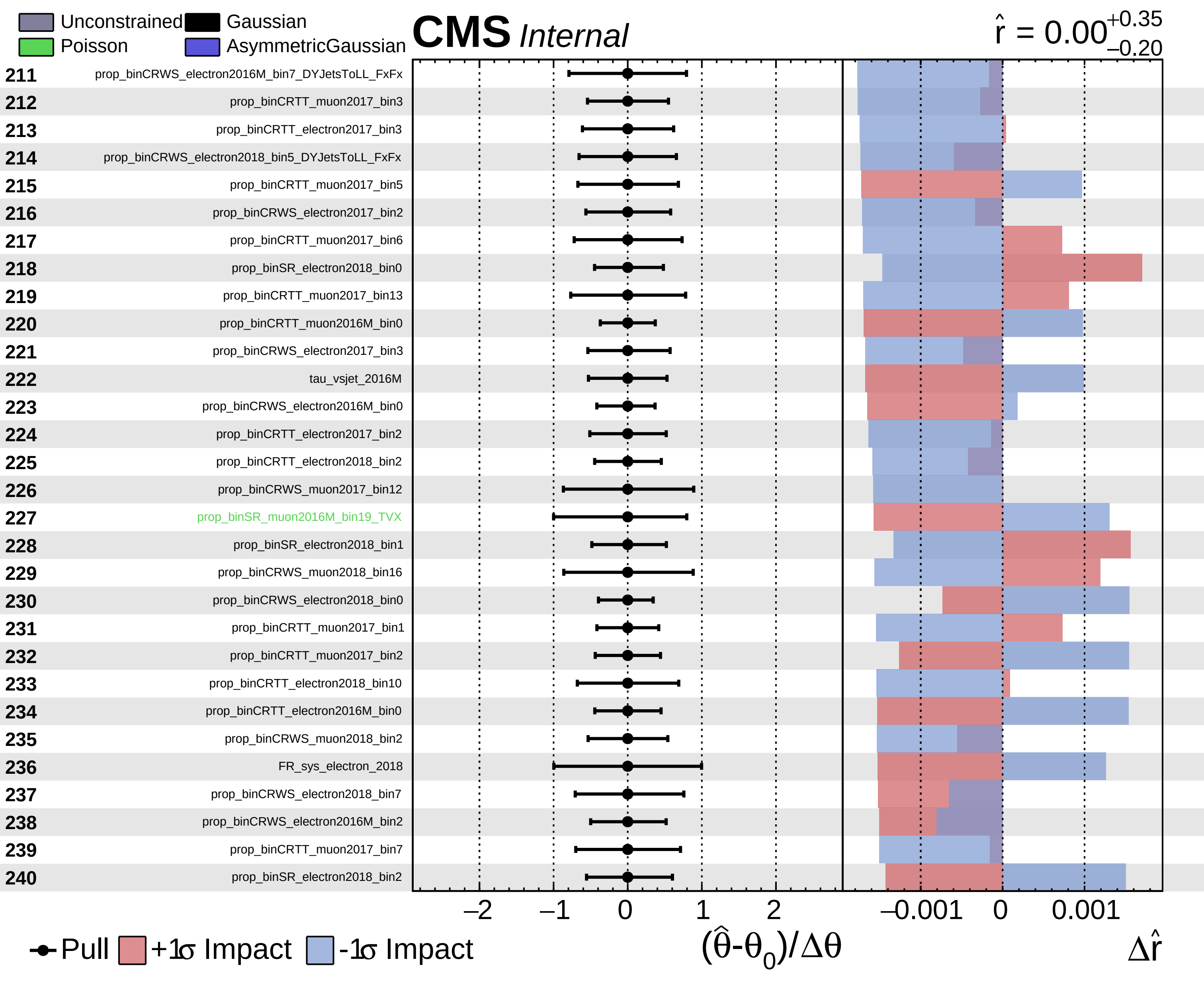

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
121 lumi_2016M
122 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
123 prop_binSR_muon2016M_bin2
124 prop_binCRWS_muon2018_bin8
125 prop_binSR_electron2016M_bin9
126 prop_binCRWS_muon2018_bin10
127 prop_binCRTT_electron2017_bin0
128 prop_binCRWS_muon2016M_bin5
129 prop_binCRWS_electron2018_bin4
130 prop_binSR_electron2018_bin4
131 prop_binCRWS_electron2016M_bin3
132 prop_binSR_electron2017_bin5
133 prop_binSR_electron2016M_bin4
134 prop_binCRWS_electron2016M_bin10
135 prop_binSR_muon2016M_bin6
136 prop_binCRWS_electron2018_bin6
137 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
138 prop_binCRWS_muon2017_bin13
139 prop_binCRWS_muon2017_bin4
140 prop_binSR_electron2016M_bin11_Fake
141 prop_binSR_electron2016M_bin14
142 prop_binCRTT_muon2018_bin0
143 prop_binSR_electron2016M_bin8
144 prop_binCRWS_electron2017_bin6
145 prop_binCRWS_electron2017_bin10
146 prop_binCRWS_muon2017_bin2
147 prop_binSR_muon2017_bin9_Fake
148 prop_binSR_muon2016M_bin3
149 prop_binCRWS_electron2016M_bin13_DYJetsToLL_FxFx
150 prop_binSR_muon2018_bin2
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



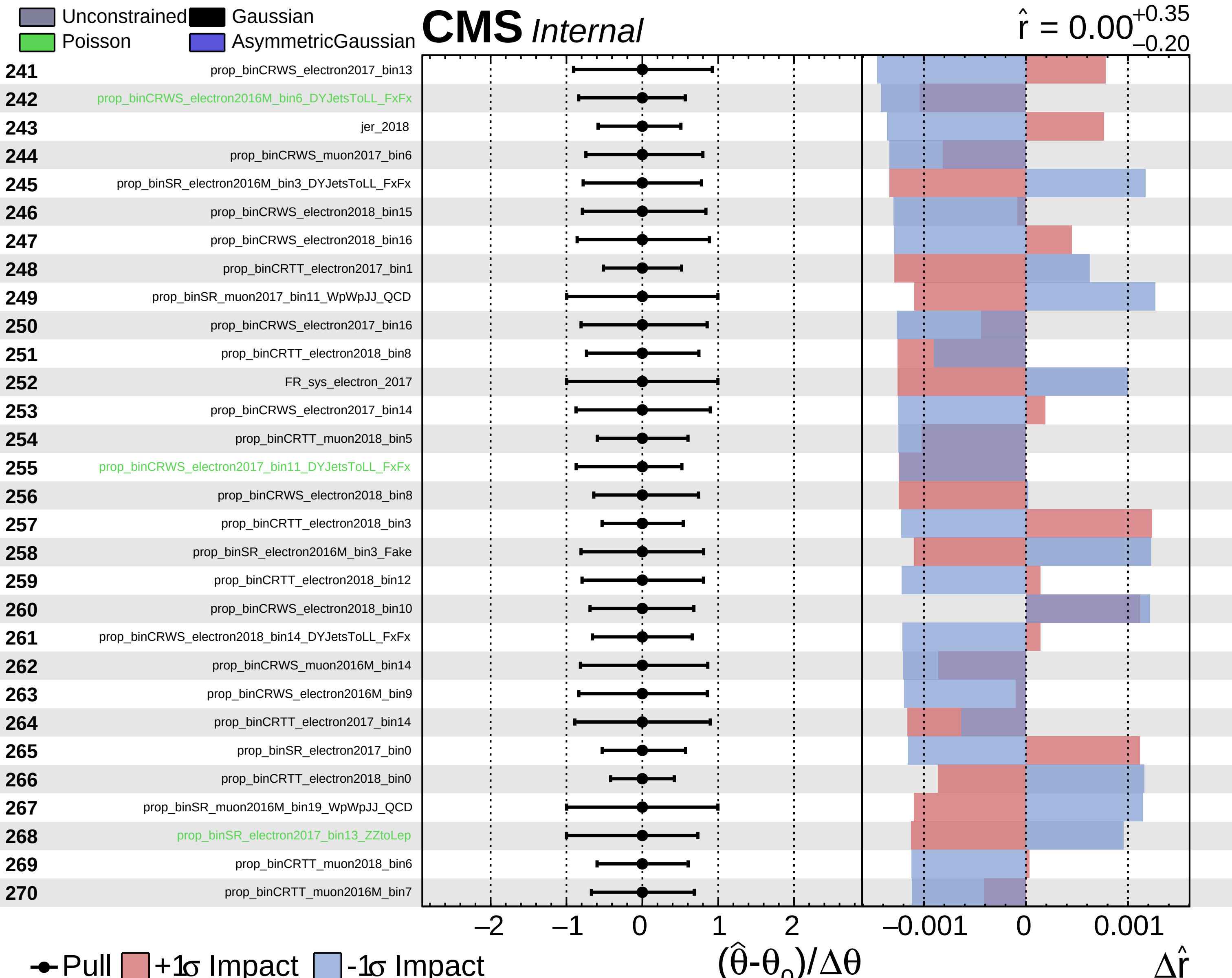
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
151 prop_binCRWS_muon2018_bin15
152 prop_binSR_electron2017_bin11
153 metUnclust
154 prop_binCRWS_muon2016M_bin0
155 prop_binCRWS_muon2018_bin6
156 prop_binSR_muon2017_bin2
157 FR_sys_muon_2016M
158 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
159 prop_binCRWS_muon2017_bin3
160 prop_binSR_electron2016M_bin12
161 prop_binCRWS_muon2018_bin17_DYJetsToLL_FxFx
162 prop_binCRWS_electron2017_bin0
163 prop_binCRWS_muon2016M_bin9
164 prop_binCRWS_muon2016M_bin3
165 prop_binCRWS_muon2017_bin17
166 prop_binSR_electron2017_bin12_TTTo2L2Nu
167 prop_binSR_electron2016M_bin13
168 prop_binCRWS_muon2018_bin4
169 prop_binCRWS_muon2017_bin11
170 prop_binCRWS_electron2018_bin3
171 jes
172 prop_binSR_muon2016M_bin5
173 prop_binSR_electron2016M_bin7_DYJetsToLL_FxFx
174 prop_binCRTT_muon2018_bin3
175 prop_binCRTT_muon2018_bin1
176 prop_binCRTT_muon2018_bin12
177 prop_binSR_electron2017_bin12_Triboson
178 prop_binSR_electron2018_bin3
179 prop_binCRWS_muon2016M_bin8
180 prop_binSR_muon2016M_bin11_ZZtoLep
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



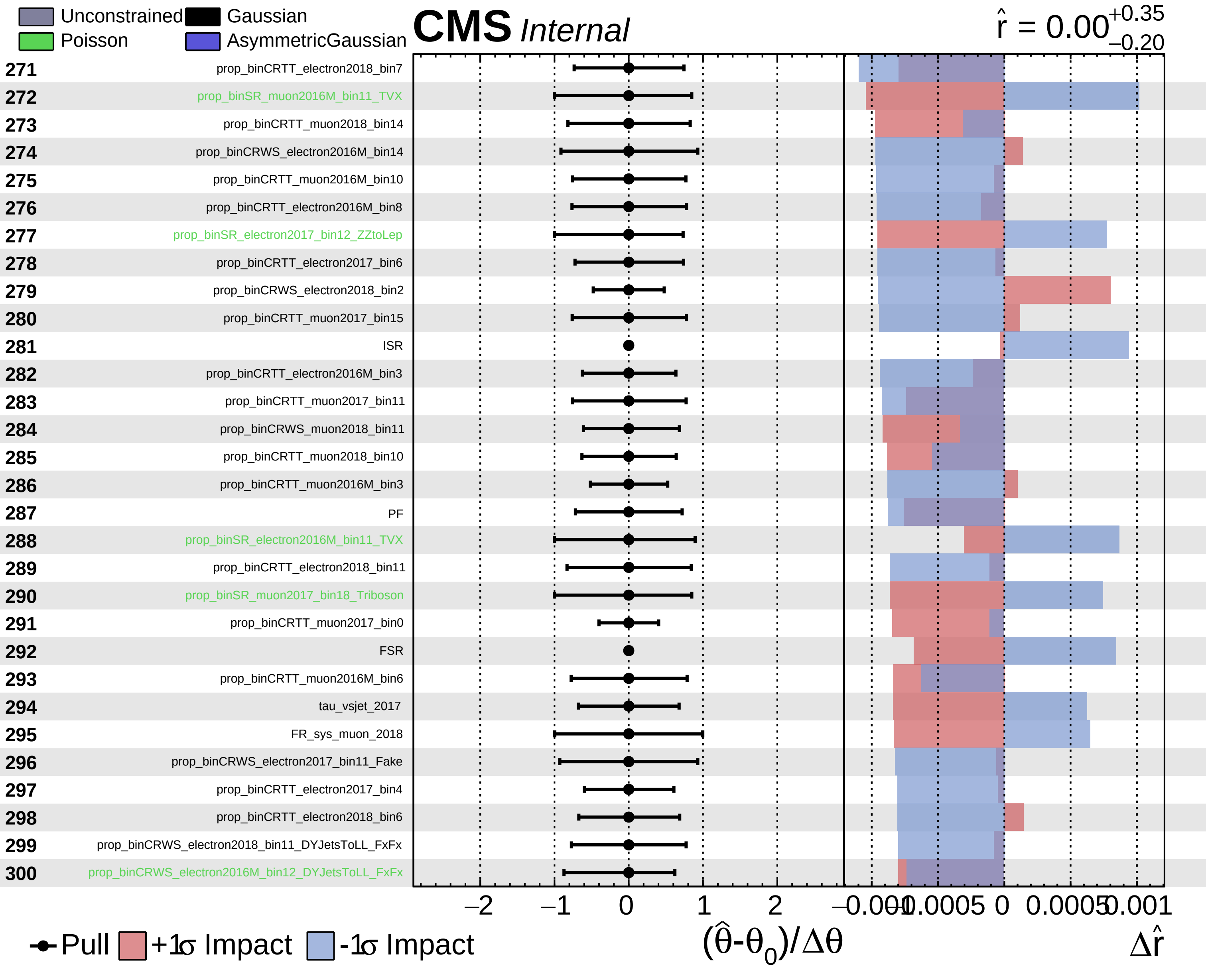
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
181 prop_binCRWS_electron2016M_bin11_DYJetsToLL_FxFx
182 prop_binCRWS_muon2016M_bin4
183 TES
184 prop_binCRTT_muon2017_bin9
185 prop_binCRWS_electron2017_bin1
186 prop_binCRWS_electron2017_bin9
187 mistag
188 prop_binSR_electron2017_bin4
189 prop_binSR_muon2018_bin0
190 prop_binCRWS_muon2016M_bin13_DYJetsToLL_FxFx
191 prop_binCRWS_muon2016M_bin6
192 prop_binSR_electron2016M_bin5
193 prop_binCRWS_muon2017_bin1
194 prop_binSR_electron2017_bin3
195 prop_binSR_muon2017_bin18_WpWpJJ_QCD
196 prop_binSR_electron2016M_bin2
197 prop_binCRWS_electron2016M_bin1
198 prop_binCRTT_muon2018_bin2
199 prop_binSR_electron2016M_bin6
200 prop_binCRWS_muon2016M_bin1
201 prop_binCRWS_electron2017_bin7
202 prop_binCRTT_muon2016M_bin1
203 prop_binSR_muon2017_bin11_TVX
204 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
205 prop_binSR_muon2016M_bin1
206 prop_binCRWS_muon2018_bin12
207 prop_binSR_muon2017_bin18_TVX
208 prop_binCRWS_muon2018_bin18_DYJetsToLL_FxFx
209 prop_binSR_electron2016M_bin1
210 prop_binSR_electron2017_bin1
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



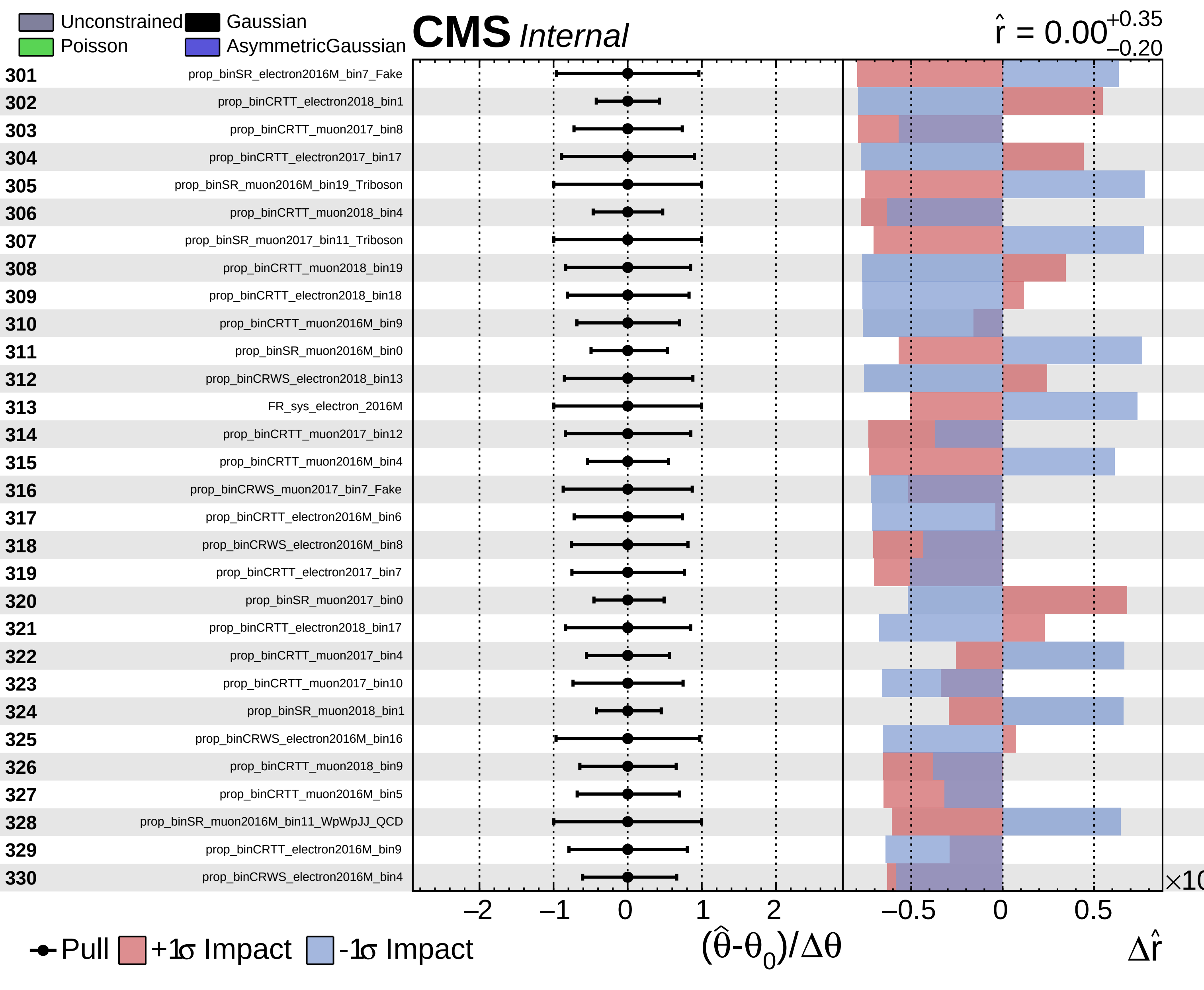
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
211 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
212 prop_binCRTT_muon2017_bin3
213 prop_binCRTT_electron2017_bin3
214 prop_binCRWS_electron2018_bin5_DYJetsToLL_FxFx
215 prop_binCRTT_muon2017_bin5
216 prop_binCRWS_electron2017_bin2
217 prop_binCRTT_muon2017_bin6
218 prop_binSR_electron2018_bin0
219 prop_binCRTT_muon2017_bin13
220 prop_binCRTT_muon2016M_bin0
221 prop_binCRWS_electron2017_bin3
222 tau_vsjet_2016M
223 prop_binCRWS_electron2016M_bin0
224 prop_binCRTT_electron2017_bin2
225 prop_binCRTT_electron2018_bin2
226 prop_binCRWS_muon2017_bin12
227 prop_binSR_muon2016M_bin19_TVX
228 prop_binSR_electron2018_bin1
229 prop_binCRWS_muon2018_bin16
230 prop_binCRWS_electron2018_bin0
231 prop_binCRTT_muon2017_bin1
232 prop_binCRTT_muon2017_bin2
233 prop_binCRTT_electron2018_bin10
234 prop_binCRTT_electron2016M_bin0
235 prop_binCRWS_muon2018_bin2
236 FR_sys_electron_2018
237 prop_binCRWS_electron2018_bin7
238 prop_binCRWS_electron2016M_bin2
239 prop_binCRTT_muon2017_bin7
240 prop_binSR_electron2018_bin2
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



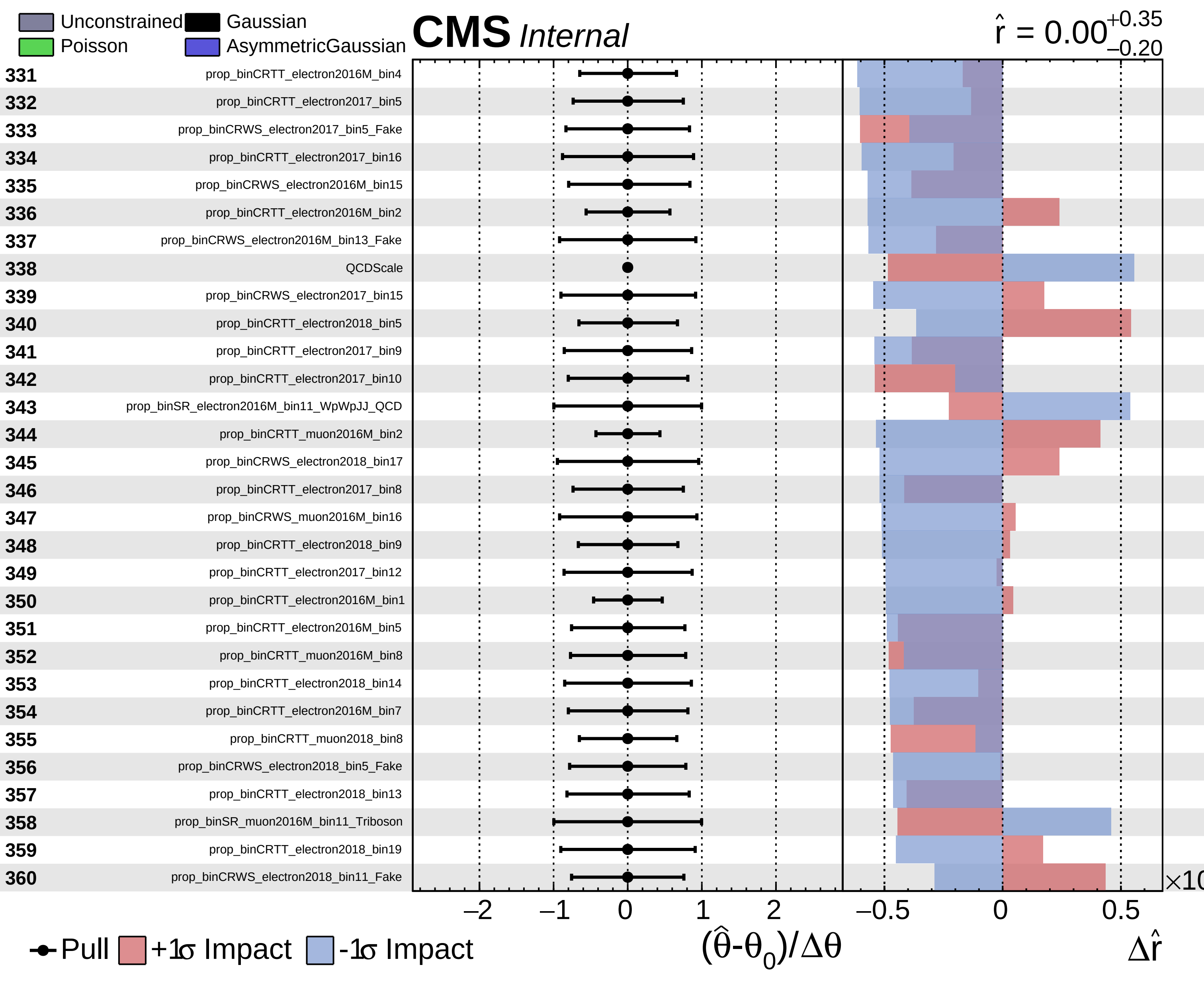
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
241 prop_binCRWS_electron2017_bin13
242 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
243 jer_2018
244 prop_binCRWS_muon2017_bin6
245 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
246 prop_binCRWS_electron2018_bin15
247 prop_binCRWS_electron2018_bin16
248 prop_binCRTT_electron2017_bin1
249 prop_binSR_muon2017_bin11_WpWpJJ_QCD
250 prop_binCRWS_electron2017_bin16
251 prop_binCRTT_electron2018_bin8
252 FR_sys_electron_2017
253 prop_binCRWS_electron2017_bin14
254 prop_binCRTT_muon2018_bin5
255 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
256 prop_binCRWS_electron2018_bin8
257 prop_binCRTT_electron2018_bin3
258 prop_binSR_electron2016M_bin3_Fake
259 prop_binCRTT_electron2018_bin12
260 prop_binCRWS_electron2018_bin10
261 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
262 prop_binCRWS_muon2016M_bin14
263 prop_binCRWS_electron2016M_bin9
264 prop_binCRTT_electron2017_bin14
265 prop_binSR_electron2017_bin0
266 prop_binCRTT_electron2018_bin0
267 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
268 prop_binSR_electron2017_bin13_ZZtoLep
269 prop_binCRTT_muon2018_bin6
270 prop_binCRTT_muon2016M_bin7
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

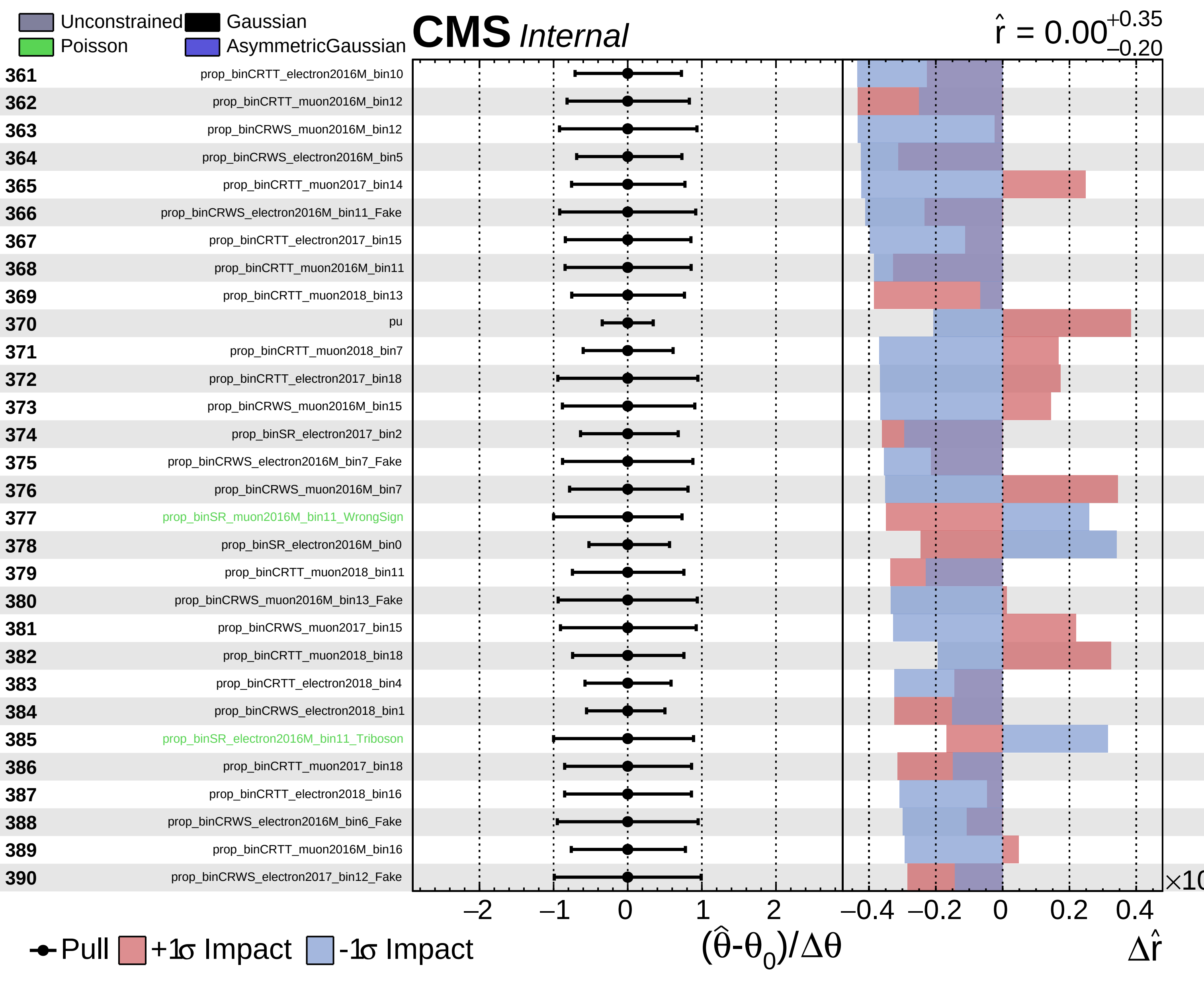


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
271 prop_binCRTT_electron2018_bin7
272 prop_binSR_muon2016M_bin11_TVX
273 prop_binCRTT_muon2018_bin14
274 prop_binCRWS_electron2016M_bin14
275 prop_binCRTT_muon2016M_bin10
276 prop_binCRTT_electron2016M_bin8
277 prop_binSR_electron2017_bin12_ZZtoLep
278 prop_binCRTT_electron2017_bin6
279 prop_binCRWS_electron2018_bin2
280 prop_binCRTT_muon2017_bin15
281 ISR
282 prop_binCRTT_electron2016M_bin3
283 prop_binCRTT_muon2017_bin11
284 prop_binCRWS_muon2018_bin11
285 prop_binCRTT_muon2018_bin10
286 prop_binCRTT_muon2016M_bin3
287 PF
288 prop_binSR_electron2016M_bin11_TVX
289 prop_binCRTT_electron2018_bin11
290 prop_binSR_muon2017_bin18_Triboson
291 prop_binCRTT_muon2017_bin0
292 FSR
293 prop_binCRTT_muon2016M_bin6
294 tau_vsjet_2017
295 FR_sys_muon_2018
296 prop_binCRWS_electron2017_bin11_Fake
297 prop_binCRTT_electron2017_bin4
298 prop_binCRTT_electron2018_bin6
299 prop_binCRWS_electron2018_bin11_DYJetsToLL_FxFx
300 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

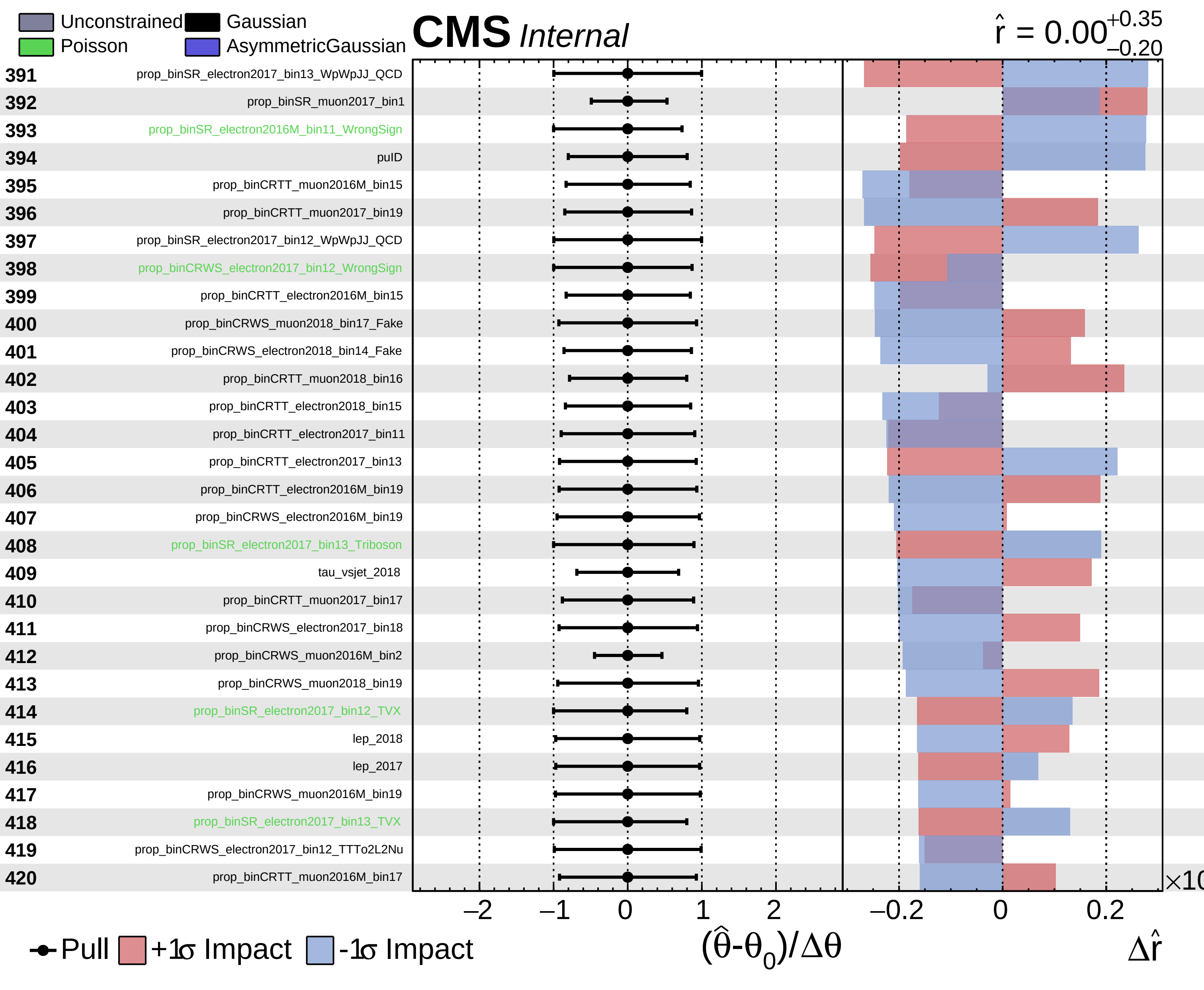


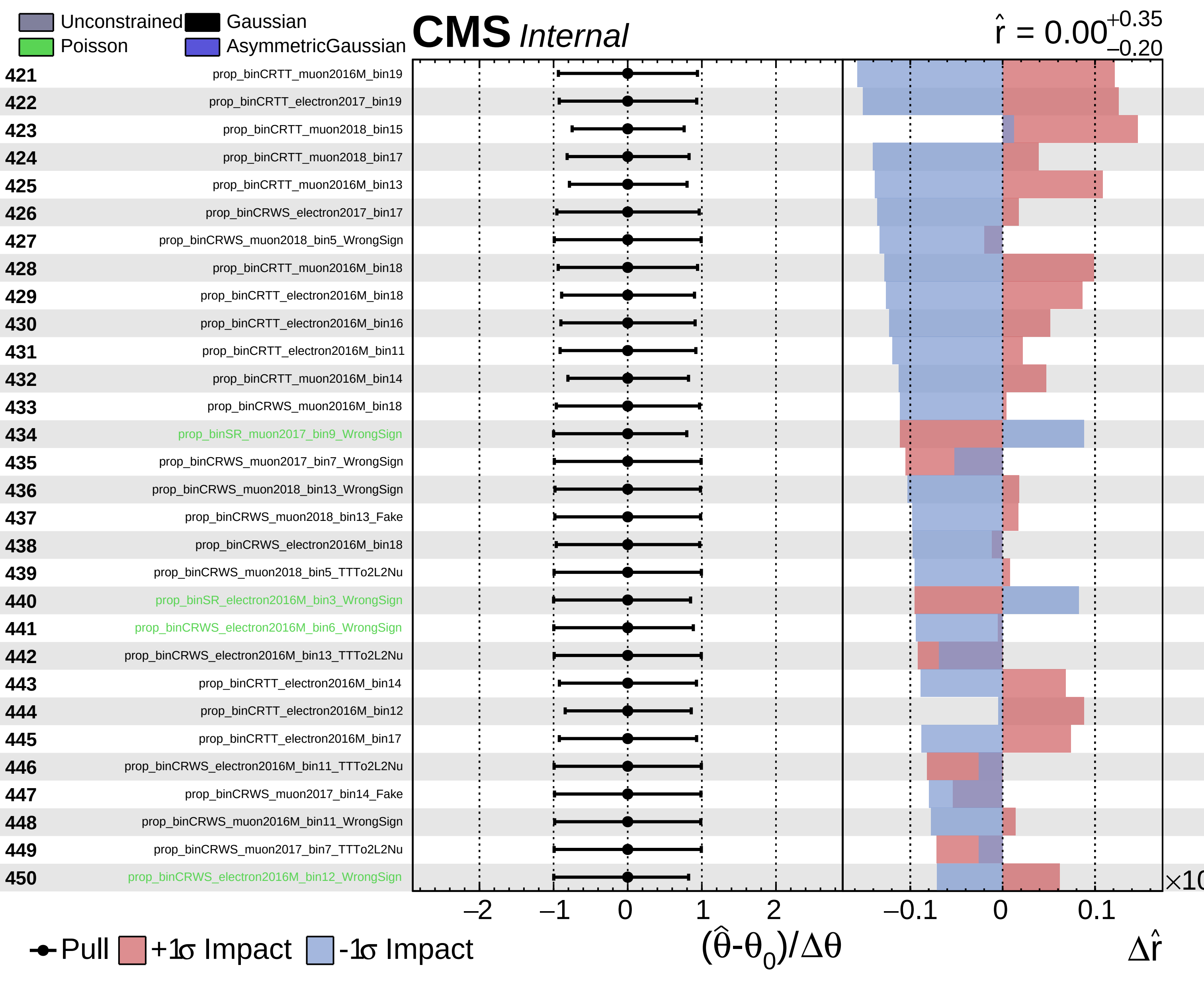
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
301 prop_binSR_electron2016M_bin7_Fake
302 prop_binCRTT_electron2018_bin1
303 prop_binCRTT_muon2017_bin8
304 prop_binCRTT_electron2017_bin17
305 prop_binSR_muon2016M_bin19_Triboson
306 prop_binCRTT_muon2018_bin4
307 prop_binSR_muon2017_bin11_Triboson
308 prop_binCRTT_muon2018_bin19
309 prop_binCRTT_electron2018_bin18
310 prop_binCRTT_muon2016M_bin9
311 prop_binSR_muon2016M_bin0
312 prop_binCRWS_electron2018_bin13
313 FR_sys_electron_2016M
314 prop_binCRTT_muon2017_bin12
315 prop_binCRTT_muon2016M_bin4
316 prop_binCRWS_muon2017_bin7_Fake
317 prop_binCRTT_electron2016M_bin6
318 prop_binCRWS_electron2016M_bin8
319 prop_binCRTT_electron2017_bin7
320 prop_binSR_muon2017_bin0
321 prop_binCRTT_electron2018_bin17
322 prop_binCRTT_muon2017_bin4
323 prop_binCRTT_muon2017_bin10
324 prop_binSR_muon2018_bin1
325 prop_binCRWS_electron2016M_bin16
326 prop_binCRTT_muon2018_bin9
327 prop_binCRTT_muon2016M_bin5
328 prop_binSR_muon2016M_bin11_WpWpJJ_QCD
329 prop_binCRTT_electron2016M_bin9
330 prop_binCRWS_electron2016M_bin4
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.35 −0.20
331 prop_binCRTT_electron2016M_bin4
332 prop_binCRTT_electron2017_bin5
333 prop_binCRWS_electron2017_bin5_Fake
334 prop_binCRTT_electron2017_bin16
335 prop_binCRWS_electron2016M_bin15
336 prop_binCRTT_electron2016M_bin2
337 prop_binCRWS_electron2016M_bin13_Fake
338 QCDScale
339 prop_binCRWS_electron2017_bin15
340 prop_binCRTT_electron2018_bin5
341 prop_binCRTT_electron2017_bin9
342 prop_binCRTT_electron2017_bin10
343 prop_binSR_electron2016M_bin11_WpWpJJ_QCD
344 prop_binCRTT_muon2016M_bin2
345 prop_binCRWS_electron2018_bin17
346 prop_binCRTT_electron2017_bin8
347 prop_binCRWS_muon2016M_bin16
348 prop_binCRTT_electron2018_bin9
349 prop_binCRTT_electron2017_bin12
350 prop_binCRTT_electron2016M_bin1
351 prop_binCRTT_electron2016M_bin5
352 prop_binCRTT_muon2016M_bin8
353 prop_binCRTT_electron2018_bin14
354 prop_binCRTT_electron2016M_bin7
355 prop_binCRTT_muon2018_bin8
356 prop_binCRWS_electron2018_bin5_Fake
357 prop_binCRTT_electron2018_bin13
358 prop_binSR_muon2016M_bin11_Triboson
359 prop_binCRTT_electron2018_bin19
360 prop_binCRWS_electron2018_bin11_Fake
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

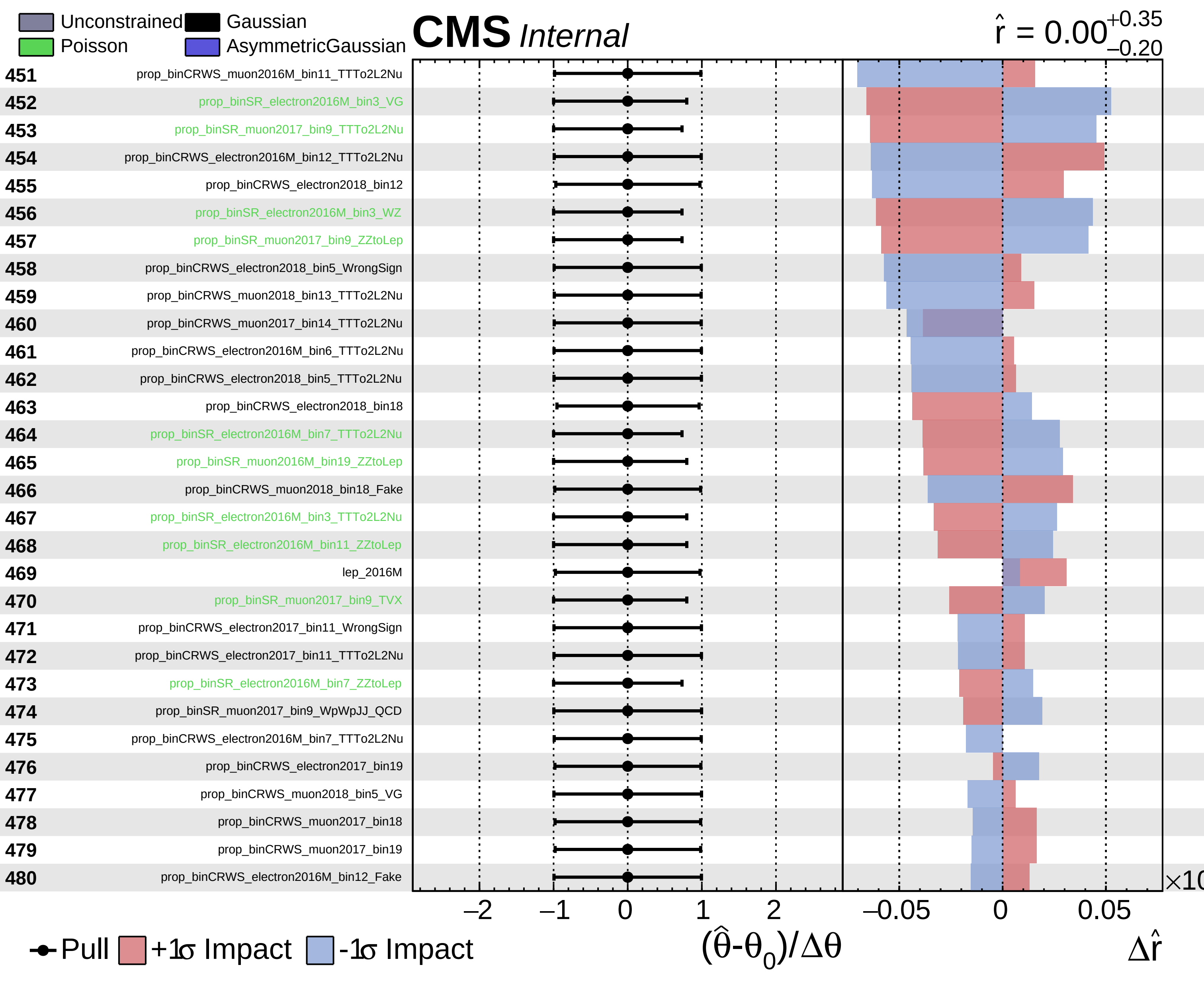


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
361 prop_binCRTT_electron2016M_bin10
362 prop_binCRTT_muon2016M_bin12
363 prop_binCRWS_muon2016M_bin12
364 prop_binCRWS_electron2016M_bin5
365 prop_binCRTT_muon2017_bin14
366 prop_binCRWS_electron2016M_bin11_Fake
367 prop_binCRTT_electron2017_bin15
368 prop_binCRTT_muon2016M_bin11
369 prop_binCRTT_muon2018_bin13
370 pu
371 prop_binCRTT_muon2018_bin7
372 prop_binCRTT_electron2017_bin18
373 prop_binCRWS_muon2016M_bin15
374 prop_binSR_electron2017_bin2
375 prop_binCRWS_electron2016M_bin7_Fake
376 prop_binCRWS_muon2016M_bin7
377 prop_binSR_muon2016M_bin11_WrongSign
378 prop_binSR_electron2016M_bin0
379 prop_binCRTT_muon2018_bin11
380 prop_binCRWS_muon2016M_bin13_Fake
381 prop_binCRWS_muon2017_bin15
382 prop_binCRTT_muon2018_bin18
383 prop_binCRTT_electron2018_bin4
384 prop_binCRWS_electron2018_bin1
385 prop_binSR_electron2016M_bin11_Triboson
386 prop_binCRTT_muon2017_bin18
387 prop_binCRTT_electron2018_bin16
388 prop_binCRWS_electron2016M_bin6_Fake
389 prop_binCRTT_muon2016M_bin16
390 prop_binCRWS_electron2017_bin12_Fake
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

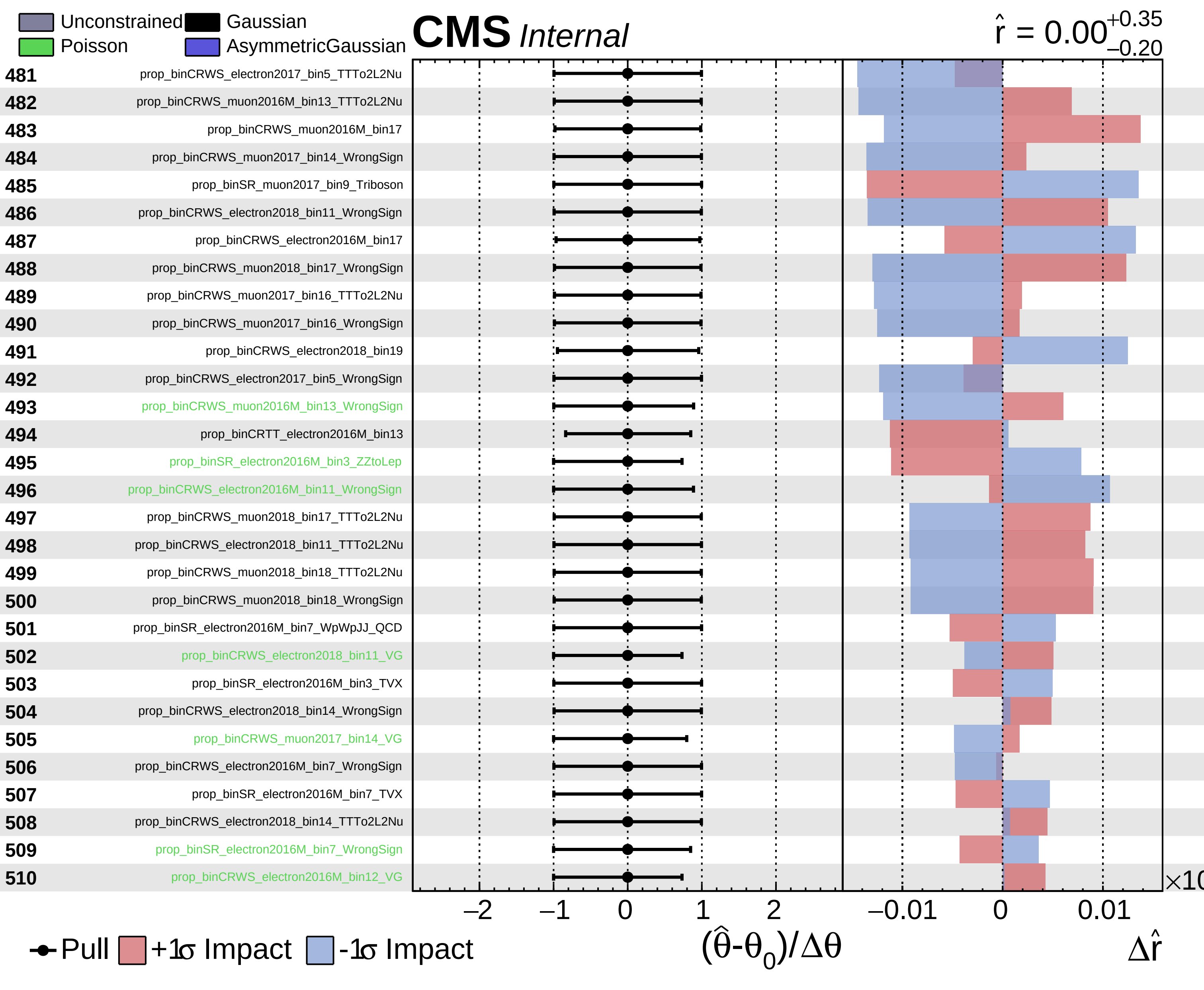

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.35 −0.20
391 prop_binSR_electron2017_bin13_WpWpJJ_QCD
392 prop_binSR_muon2017_bin1
393 prop_binSR_electron2016M_bin11_WrongSign
394 puID
395 prop_binCRTT_muon2016M_bin15
396 prop_binCRTT_muon2017_bin19
397 prop_binSR_electron2017_bin12_WpWpJJ_QCD
398 prop_binCRWS_electron2017_bin12_WrongSign
399 prop_binCRTT_electron2016M_bin15
400 prop_binCRWS_muon2018_bin17_Fake
401 prop_binCRWS_electron2018_bin14_Fake
402 prop_binCRTT_muon2018_bin16
403 prop_binCRTT_electron2018_bin15
404 prop_binCRTT_electron2017_bin11
405 prop_binCRTT_electron2017_bin13
406 prop_binCRTT_electron2016M_bin19
407 prop_binCRWS_electron2016M_bin19
408 prop_binSR_electron2017_bin13_Triboson
409 tau_vsjet_2018
410 prop_binCRTT_muon2017_bin17
411 prop_binCRWS_electron2017_bin18
412 prop_binCRWS_muon2016M_bin2
413 prop_binCRWS_muon2018_bin19
414 prop_binSR_electron2017_bin12_TVX
415 lep_2018
416 lep_2017
417 prop_binCRWS_muon2016M_bin19
418 prop_binSR_electron2017_bin13_TVX
419 prop_binCRWS_electron2017_bin12_TTTo2L2Nu
420 prop_binCRTT_muon2016M_bin17
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

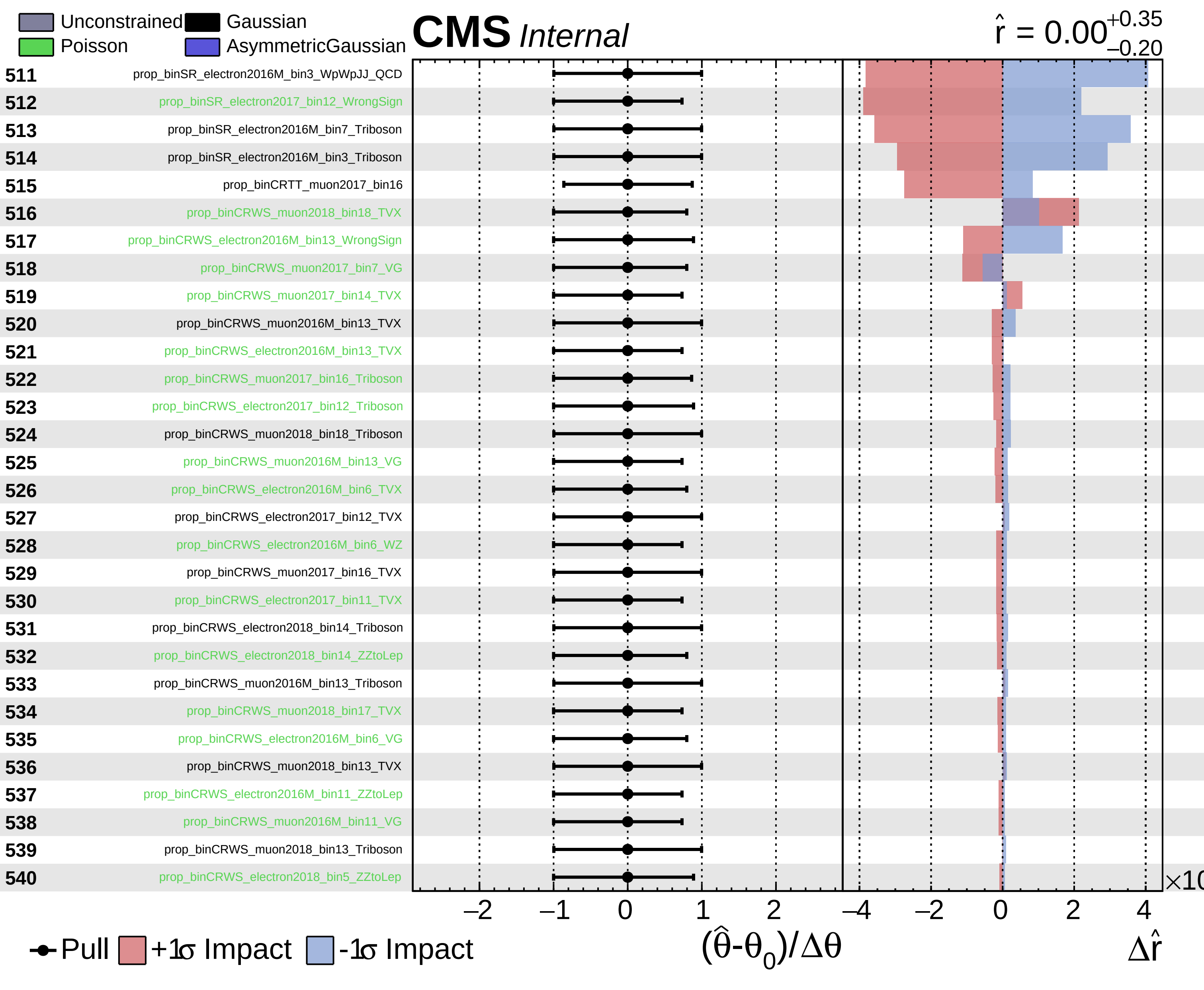


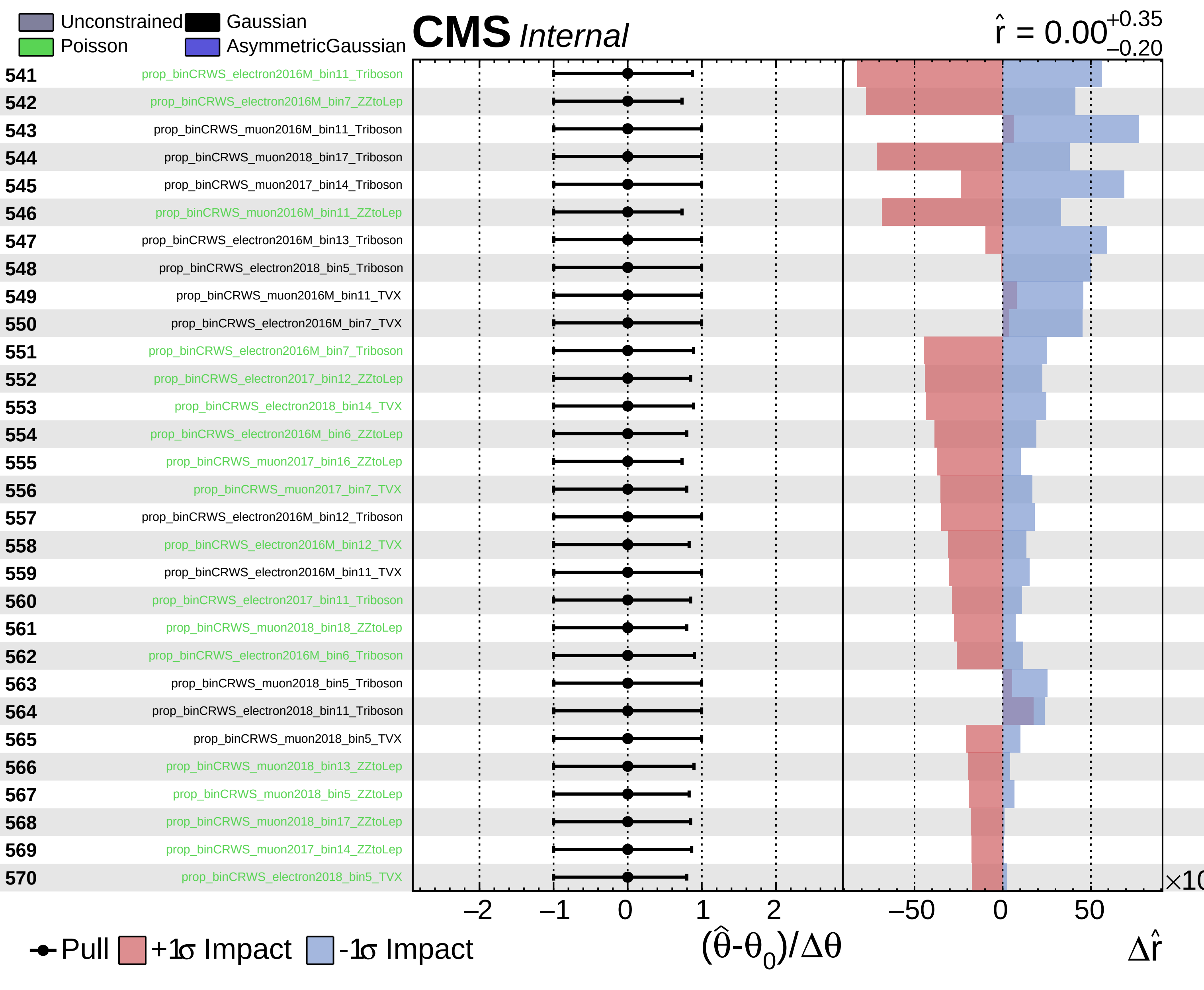
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
421 prop_binCRTT_muon2016M_bin19
422 prop_binCRTT_electron2017_bin19
423 prop_binCRTT_muon2018_bin15
424 prop_binCRTT_muon2018_bin17
425 prop_binCRTT_muon2016M_bin13
426 prop_binCRWS_electron2017_bin17
427 prop_binCRWS_muon2018_bin5_WrongSign
428 prop_binCRTT_muon2016M_bin18
429 prop_binCRTT_electron2016M_bin18
430 prop_binCRTT_electron2016M_bin16
431 prop_binCRTT_electron2016M_bin11
432 prop_binCRTT_muon2016M_bin14
433 prop_binCRWS_muon2016M_bin18
434 prop_binSR_muon2017_bin9_WrongSign
435 prop_binCRWS_muon2017_bin7_WrongSign
436 prop_binCRWS_muon2018_bin13_WrongSign
437 prop_binCRWS_muon2018_bin13_Fake
438 prop_binCRWS_electron2016M_bin18
439 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
440 prop_binSR_electron2016M_bin3_WrongSign
441 prop_binCRWS_electron2016M_bin6_WrongSign
442 prop_binCRWS_electron2016M_bin13_TTTo2L2Nu
443 prop_binCRTT_electron2016M_bin14
444 prop_binCRTT_electron2016M_bin12
445 prop_binCRTT_electron2016M_bin17
446 prop_binCRWS_electron2016M_bin11_TTTo2L2Nu
447 prop_binCRWS_muon2017_bin14_Fake
448 prop_binCRWS_muon2016M_bin11_WrongSign
449 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
450 prop_binCRWS_electron2016M_bin12_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



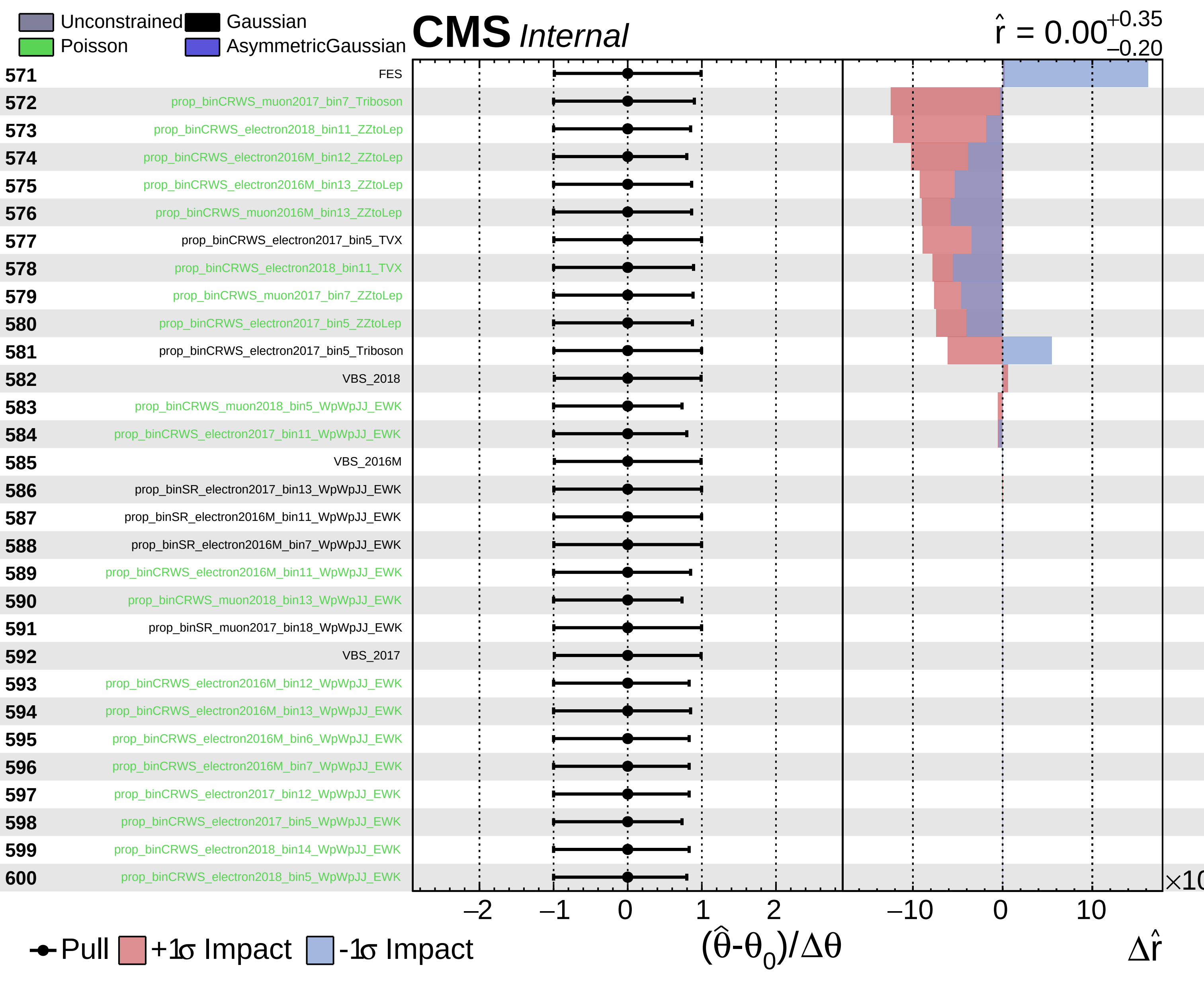
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
451 prop_binCRWS_muon2016M_bin11_TTTo2L2Nu
452 prop_binSR_electron2016M_bin3_VG
453 prop_binSR_muon2017_bin9_TTTo2L2Nu
454 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
455 prop_binCRWS_electron2018_bin12
456 prop_binSR_electron2016M_bin3_WZ
457 prop_binSR_muon2017_bin9_ZZtoLep
458 prop_binCRWS_electron2018_bin5_WrongSign
459 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
460 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
461 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
462 prop_binCRWS_electron2018_bin5_TTTo2L2Nu
463 prop_binCRWS_electron2018_bin18
464 prop_binSR_electron2016M_bin7_TTTo2L2Nu
465 prop_binSR_muon2016M_bin19_ZZtoLep
466 prop_binCRWS_muon2018_bin18_Fake
467 prop_binSR_electron2016M_bin3_TTTo2L2Nu
468 prop_binSR_electron2016M_bin11_ZZtoLep
469 lep_2016M
470 prop_binSR_muon2017_bin9_TVX
471 prop_binCRWS_electron2017_bin11_WrongSign
472 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
473 prop_binSR_electron2016M_bin7_ZZtoLep
474 prop_binSR_muon2017_bin9_WpWpJJ_QCD
475 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
476 prop_binCRWS_electron2017_bin19
477 prop_binCRWS_muon2018_bin5_VG
478 prop_binCRWS_muon2017_bin18
479 prop_binCRWS_muon2017_bin19
480 prop_binCRWS_electron2016M_bin12_Fake
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

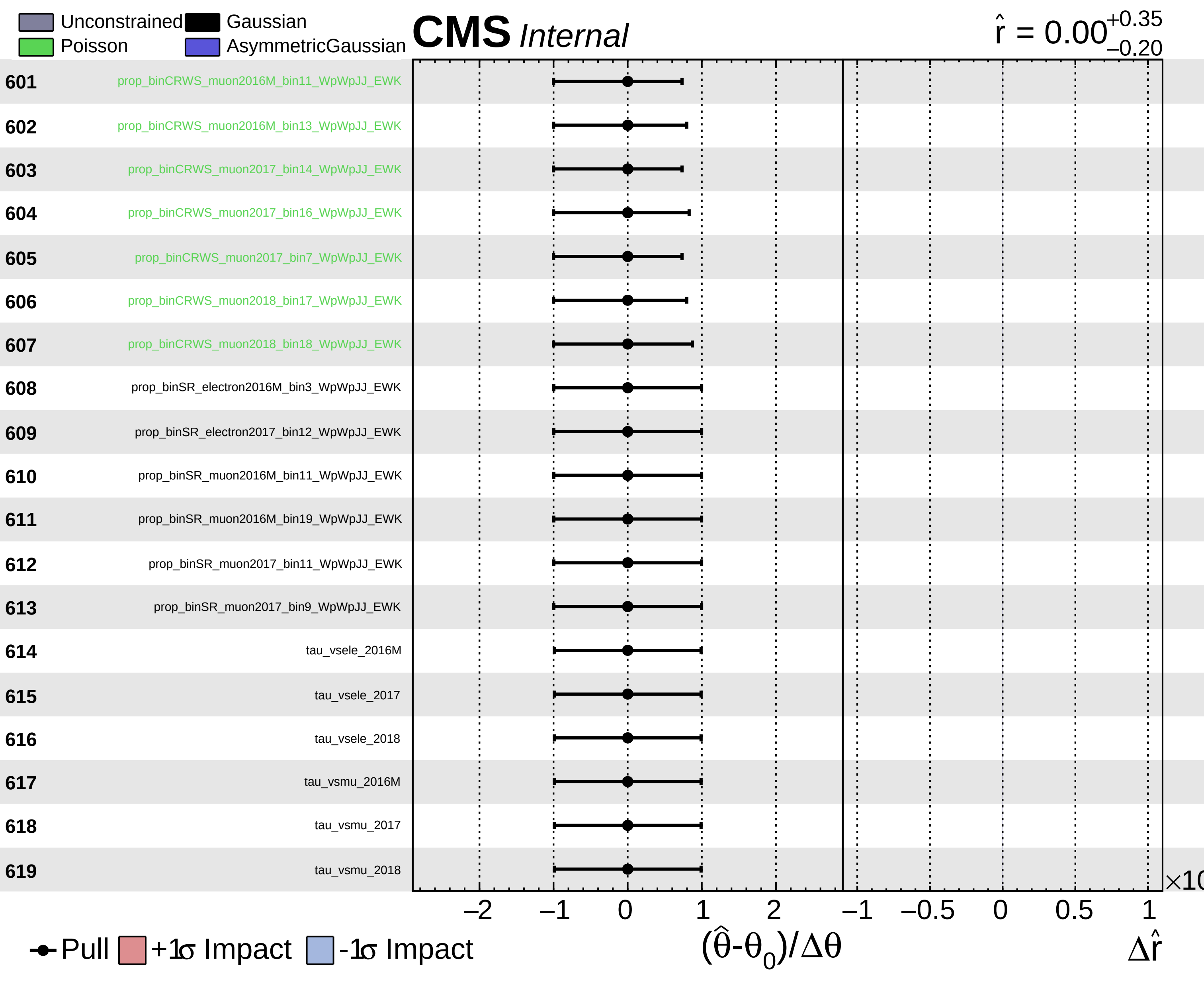

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.35 −0.20
481 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
482 prop_binCRWS_muon2016M_bin13_TTTo2L2Nu
483 prop_binCRWS_muon2016M_bin17
484 prop_binCRWS_muon2017_bin14_WrongSign
485 prop_binSR_muon2017_bin9_Triboson
486 prop_binCRWS_electron2018_bin11_WrongSign
487 prop_binCRWS_electron2016M_bin17
488 prop_binCRWS_muon2018_bin17_WrongSign
489 prop_binCRWS_muon2017_bin16_TTTo2L2Nu
490 prop_binCRWS_muon2017_bin16_WrongSign
491 prop_binCRWS_electron2018_bin19
492 prop_binCRWS_electron2017_bin5_WrongSign
493 prop_binCRWS_muon2016M_bin13_WrongSign
494 prop_binCRTT_electron2016M_bin13
495 prop_binSR_electron2016M_bin3_ZZtoLep
496 prop_binCRWS_electron2016M_bin11_WrongSign
497 prop_binCRWS_muon2018_bin17_TTTo2L2Nu
498 prop_binCRWS_electron2018_bin11_TTTo2L2Nu
499 prop_binCRWS_muon2018_bin18_TTTo2L2Nu
500 prop_binCRWS_muon2018_bin18_WrongSign
501 prop_binSR_electron2016M_bin7_WpWpJJ_QCD
502 prop_binCRWS_electron2018_bin11_VG
503 prop_binSR_electron2016M_bin3_TVX
504 prop_binCRWS_electron2018_bin14_WrongSign
505 prop_binCRWS_muon2017_bin14_VG
506 prop_binCRWS_electron2016M_bin7_WrongSign
507 prop_binSR_electron2016M_bin7_TVX
508 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
509 prop_binSR_electron2016M_bin7_WrongSign
510 prop_binCRWS_electron2016M_bin12_VG
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.35 −0.20
511 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
512 prop_binSR_electron2017_bin12_WrongSign
513 prop_binSR_electron2016M_bin7_Triboson
514 prop_binSR_electron2016M_bin3_Triboson
515 prop_binCRTT_muon2017_bin16
516 prop_binCRWS_muon2018_bin18_TVX
517 prop_binCRWS_electron2016M_bin13_WrongSign
518 prop_binCRWS_muon2017_bin7_VG
519 prop_binCRWS_muon2017_bin14_TVX
520 prop_binCRWS_muon2016M_bin13_TVX
521 prop_binCRWS_electron2016M_bin13_TVX
522 prop_binCRWS_muon2017_bin16_Triboson
523 prop_binCRWS_electron2017_bin12_Triboson
524 prop_binCRWS_muon2018_bin18_Triboson
525 prop_binCRWS_muon2016M_bin13_VG
526 prop_binCRWS_electron2016M_bin6_TVX
527 prop_binCRWS_electron2017_bin12_TVX
528 prop_binCRWS_electron2016M_bin6_WZ
529 prop_binCRWS_muon2017_bin16_TVX
530 prop_binCRWS_electron2017_bin11_TVX
531 prop_binCRWS_electron2018_bin14_Triboson
532 prop_binCRWS_electron2018_bin14_ZZtoLep
533 prop_binCRWS_muon2016M_bin13_Triboson
534 prop_binCRWS_muon2018_bin17_TVX
535 prop_binCRWS_electron2016M_bin6_VG
536 prop_binCRWS_muon2018_bin13_TVX
537 prop_binCRWS_electron2016M_bin11_ZZtoLep
538 prop_binCRWS_muon2016M_bin11_VG
539 prop_binCRWS_muon2018_bin13_Triboson
540 prop_binCRWS_electron2018_bin5_ZZtoLep
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
541 prop_binCRWS_electron2016M_bin11_Triboson
542 prop_binCRWS_electron2016M_bin7_ZZtoLep
543 prop_binCRWS_muon2016M_bin11_Triboson
544 prop_binCRWS_muon2018_bin17_Triboson
545 prop_binCRWS_muon2017_bin14_Triboson
546 prop_binCRWS_muon2016M_bin11_ZZtoLep
547 prop_binCRWS_electron2016M_bin13_Triboson
548 prop_binCRWS_electron2018_bin5_Triboson
549 prop_binCRWS_muon2016M_bin11_TVX
550 prop_binCRWS_electron2016M_bin7_TVX
551 prop_binCRWS_electron2016M_bin7_Triboson
552 prop_binCRWS_electron2017_bin12_ZZtoLep
553 prop_binCRWS_electron2018_bin14_TVX
554 prop_binCRWS_electron2016M_bin6_ZZtoLep
555 prop_binCRWS_muon2017_bin16_ZZtoLep
556 prop_binCRWS_muon2017_bin7_TVX
557 prop_binCRWS_electron2016M_bin12_Triboson
558 prop_binCRWS_electron2016M_bin12_TVX
559 prop_binCRWS_electron2016M_bin11_TVX
560 prop_binCRWS_electron2017_bin11_Triboson
561 prop_binCRWS_muon2018_bin18_ZZtoLep
562 prop_binCRWS_electron2016M_bin6_Triboson
563 prop_binCRWS_muon2018_bin5_Triboson
564 prop_binCRWS_electron2018_bin11_Triboson
565 prop_binCRWS_muon2018_bin5_TVX
566 prop_binCRWS_muon2018_bin13_ZZtoLep
567 prop_binCRWS_muon2018_bin5_ZZtoLep
568 prop_binCRWS_muon2018_bin17_ZZtoLep
569 prop_binCRWS_muon2017_bin14_ZZtoLep
570 prop_binCRWS_electron2018_bin5_TVX
−2 −1 0 1 2
−50 0 50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
571 FES
572 prop_binCRWS_muon2017_bin7_Triboson
573 prop_binCRWS_electron2018_bin11_ZZtoLep
574 prop_binCRWS_electron2016M_bin12_ZZtoLep
575 prop_binCRWS_electron2016M_bin13_ZZtoLep
576 prop_binCRWS_muon2016M_bin13_ZZtoLep
577 prop_binCRWS_electron2017_bin5_TVX
578 prop_binCRWS_electron2018_bin11_TVX
579 prop_binCRWS_muon2017_bin7_ZZtoLep
580 prop_binCRWS_electron2017_bin5_ZZtoLep
581 prop_binCRWS_electron2017_bin5_Triboson
582 VBS_2018
583 prop_binCRWS_muon2018_bin5_WpWpJJ_EWK
584 prop_binCRWS_electron2017_bin11_WpWpJJ_EWK
585 VBS_2016M
586 prop_binSR_electron2017_bin13_WpWpJJ_EWK
587 prop_binSR_electron2016M_bin11_WpWpJJ_EWK
588 prop_binSR_electron2016M_bin7_WpWpJJ_EWK
589 prop_binCRWS_electron2016M_bin11_WpWpJJ_EWK
590 prop_binCRWS_muon2018_bin13_WpWpJJ_EWK
591 prop_binSR_muon2017_bin18_WpWpJJ_EWK
592 VBS_2017
593 prop_binCRWS_electron2016M_bin12_WpWpJJ_EWK
594 prop_binCRWS_electron2016M_bin13_WpWpJJ_EWK
595 prop_binCRWS_electron2016M_bin6_WpWpJJ_EWK
596 prop_binCRWS_electron2016M_bin7_WpWpJJ_EWK
597 prop_binCRWS_electron2017_bin12_WpWpJJ_EWK
598 prop_binCRWS_electron2017_bin5_WpWpJJ_EWK
599 prop_binCRWS_electron2018_bin14_WpWpJJ_EWK
600 prop_binCRWS_electron2018_bin5_WpWpJJ_EWK
−2 −1 0 1 2
−10 0 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.20}$
601 prop_binCRWS_muon2016M_bin11_WpWpJJ_EWK
602 prop_binCRWS_muon2016M_bin13_WpWpJJ_EWK
603 prop_binCRWS_muon2017_bin14_WpWpJJ_EWK
604 prop_binCRWS_muon2017_bin16_WpWpJJ_EWK
605 prop_binCRWS_muon2017_bin7_WpWpJJ_EWK
606 prop_binCRWS_muon2018_bin17_WpWpJJ_EWK
607 prop_binCRWS_muon2018_bin18_WpWpJJ_EWK
608 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
609 prop_binSR_electron2017_bin12_WpWpJJ_EWK
610 prop_binSR_muon2016M_bin11_WpWpJJ_EWK
611 prop_binSR_muon2016M_bin19_WpWpJJ_EWK
612 prop_binSR_muon2017_bin11_WpWpJJ_EWK
613 prop_binSR_muon2017_bin9_WpWpJJ_EWK
614 tau_vsele_2016M
615 tau_vsele_2017
616 tau_vsele_2018
617 tau_vsmu_2016M
618 tau_vsmu_2017
619 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$